



集成电路设计工程 2025秋

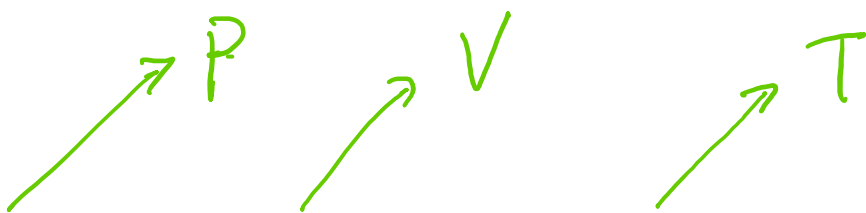
Lecture 8 电路可靠性

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清华大学

2025年11月11日

Outline



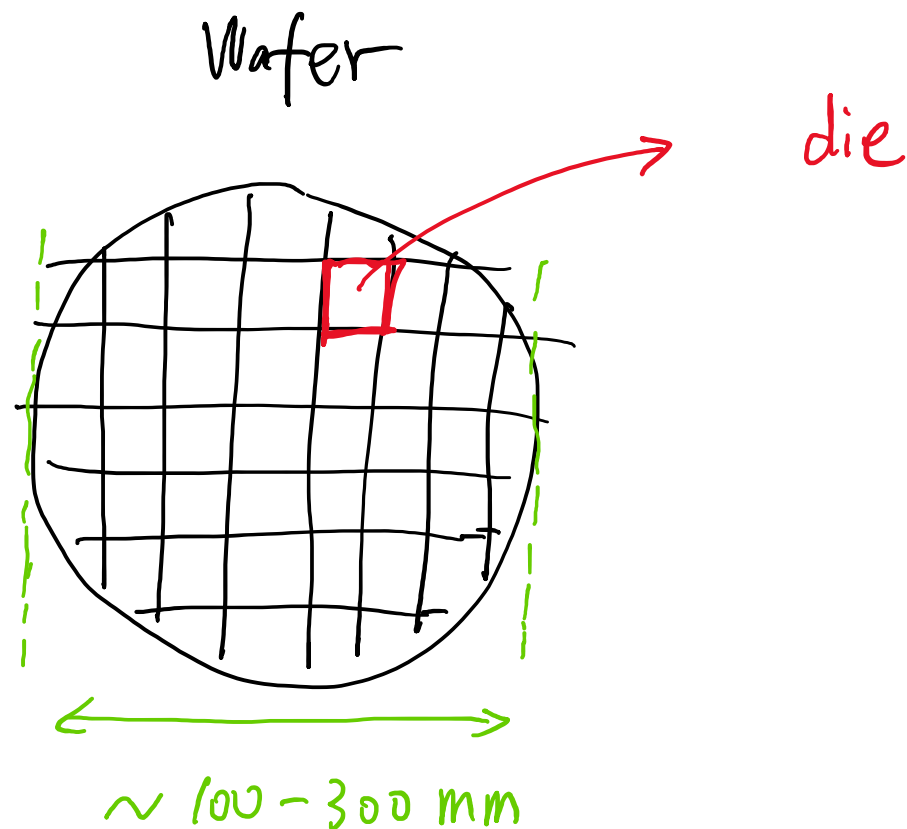
- Process, voltage, temperature variations
- Local mismatch
- Monte Carlo simulation



Process Variation



Process Variation (across different wafers/chips)



Agencies like MOSIS combine different chip designs with different sizes on a wafer for fabrication.

Called multiproject wafer
(MPW)

Process Variation

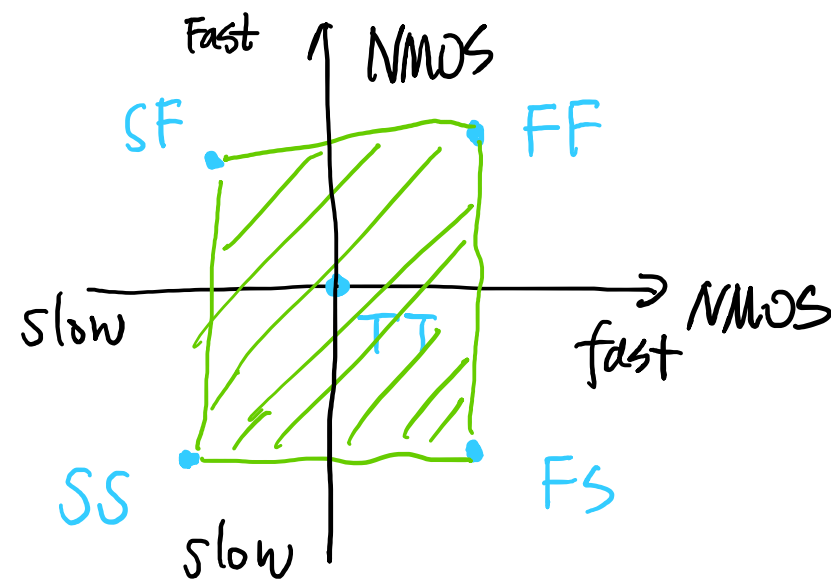


Corners : Typical / Normal
(T) (N)

Fast
(F)

Slow
(S)

For transistors:



Process Variation



Simulate extreme corners:

	NMOS	PMOS	Temp.	VDD	R	C
typical	T	T	27°	1V	"	"
Fast	F	F	75°	1.1V (+10%)	"	"
Slow	S	S	-25°	0.9V (-10%)	"	"

Baseline !!

Process Variation



More checks for better reliability:

Example of digital verification:

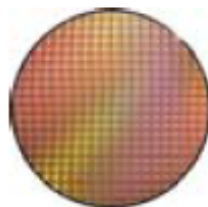
Corner					Purpose
nMOS	pMOS	Wire	V_{DD}	Temp	
T	T	T	S	S	Timing specifications (binned parts)
S	S	S	S	S	Timing specifications (conservative)
F	F	F	F	F	Race conditions, hold time constraints, pulse collapse, noise
S	S	?	F	S	Dynamic power
F	F	F	F	S	Subthreshold leakage noise and power, overall noise analysis
S	S	F	S	S	Races of gates against wires
F	F	S	F	F	Races of wires against gates
S	F	T	F	F	Pseudo-nMOS and ratioed circuits noise margins, memory read/write, race of pMOS against nMOS
F	S	T	F	F	Ratioed circuits, memory read/write, race of nMOS against pMOS

Process Variation



Wafer made yesterday

All NMOS are “slow”
All PMOS are “nominal”
All R are nominal
All C are “fast”



Wafer made today

All NMOS are “fast”
All PMOS are “fast”
All R are nominal
All C are “slow”

Parameter	“Slow”	“Nominal”	“Fast”
V_t	0.65V	0.5V	0.35V
μC_{ox} (NMOS)	40 $\mu A/V^2$	50 $\mu A/V^2$	60 $\mu A/V^2$
μC_{ox} (PMOS)	20 $\mu A/V^2$	25 $\mu A/V^2$	30 $\mu A/V^2$
R_{poly2}	60 $\Omega/\square$	50 $\Omega/\square$	40 $\Omega/\square$
R_{nwell}	1.4 k $\Omega/\square$	1 k $\Omega/\square$	0.6 k $\Omega/\square$
$C_{poly-poly2}$	1.15 fF/ μm^2	1 fF/ μm^2	0.85 fF/ μm^2

Doping densities

N_A and N_D variation
due to dopings
(vertical dimensions: vary
the most)

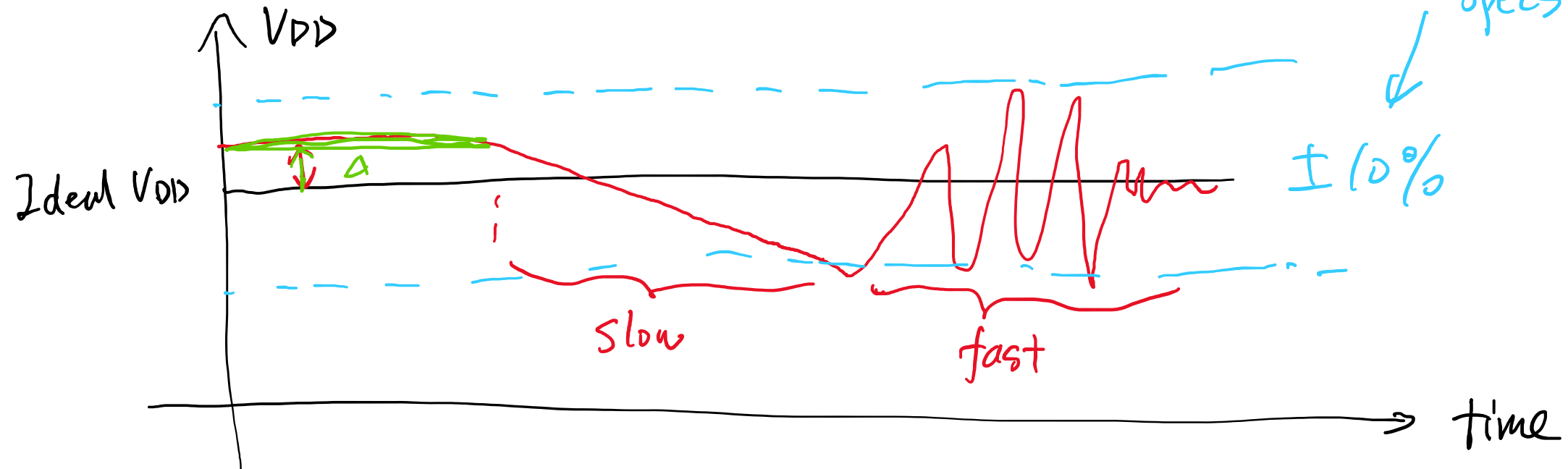
* lateral dimensions
are better controlled

[Stanford, B. Murmann]

Supply Voltage Variation



- ① Regulator error (Δ)
- ② On-chip droop due to circuit operations



Temperature Variation



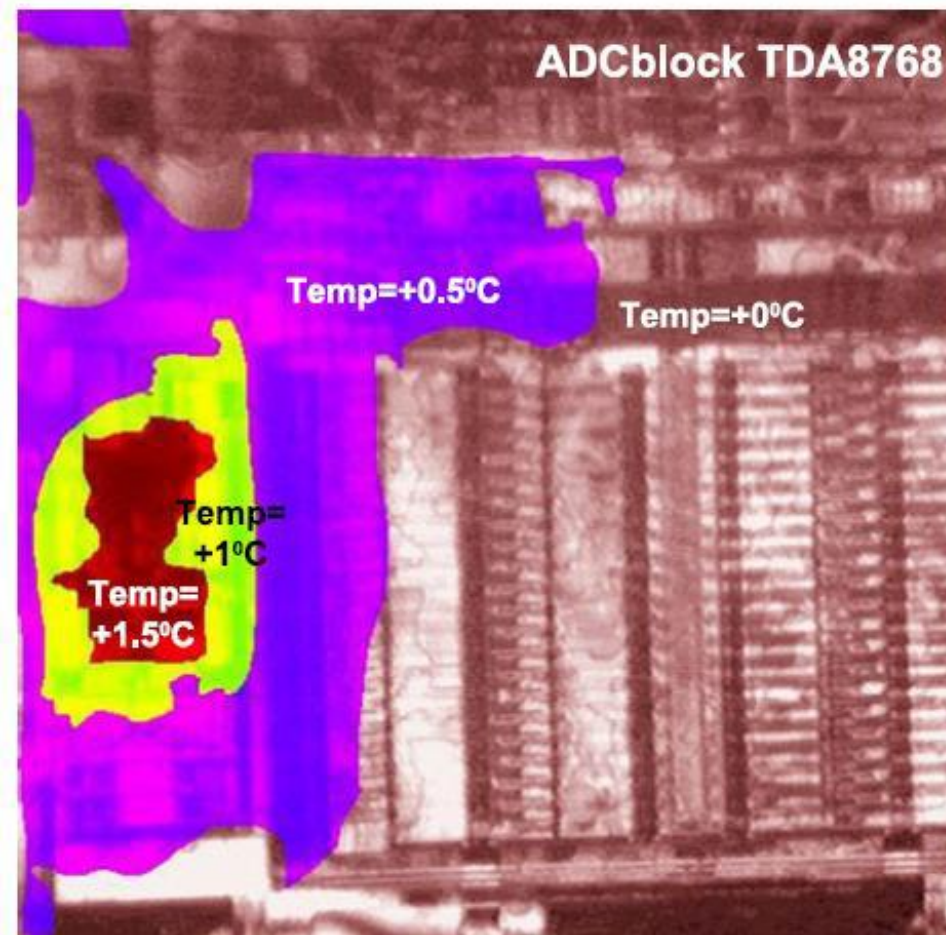
Temperature Variation

① Environment temperature change.

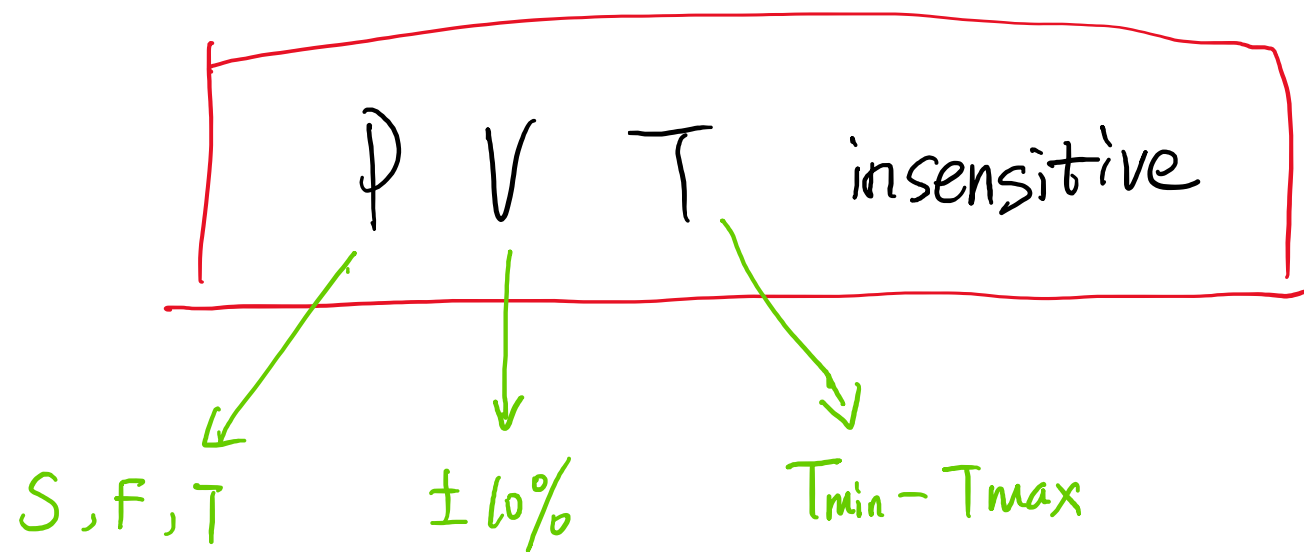
② Circuit-related heating
(P_{diss}) $\Rightarrow$

-25°C to $+75^{\circ}\text{C}$

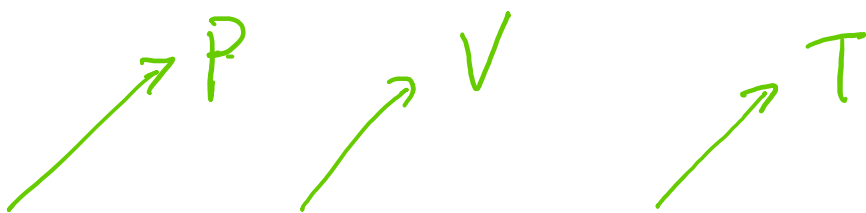
-40°C to $+120^{\circ}\text{C}$



Design Goal



Outline

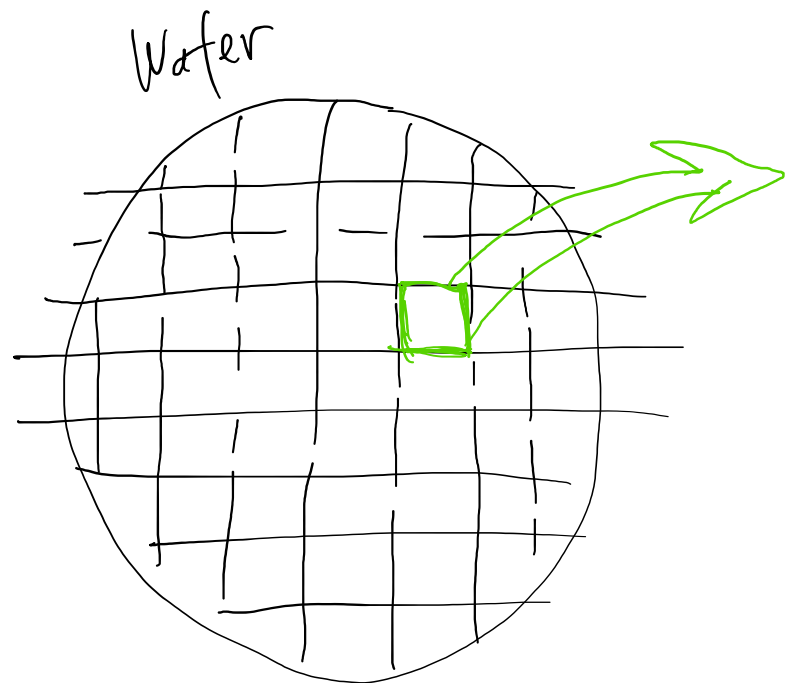


- Process, voltage, temperature variations
- Local mismatch
- Monte Carlo simulation
- Mismatch errors and mitigation techniques
- Layout-related mismatch

Component Mismatch



Mismatch - Differences between identical devices on the same die.



M1

M2

$$\begin{cases} \Delta V_{th} = V_{th,1} - V_{th,2} \\ \Delta \beta = \left(\mu_n C_{ox} \frac{W}{L} \right)_1 - \left(\mu_n C_{ox} \frac{W}{L} \right)_2 \end{cases}$$

C1

C2

$$\Delta C = C_1 - C_2$$

R1

R2

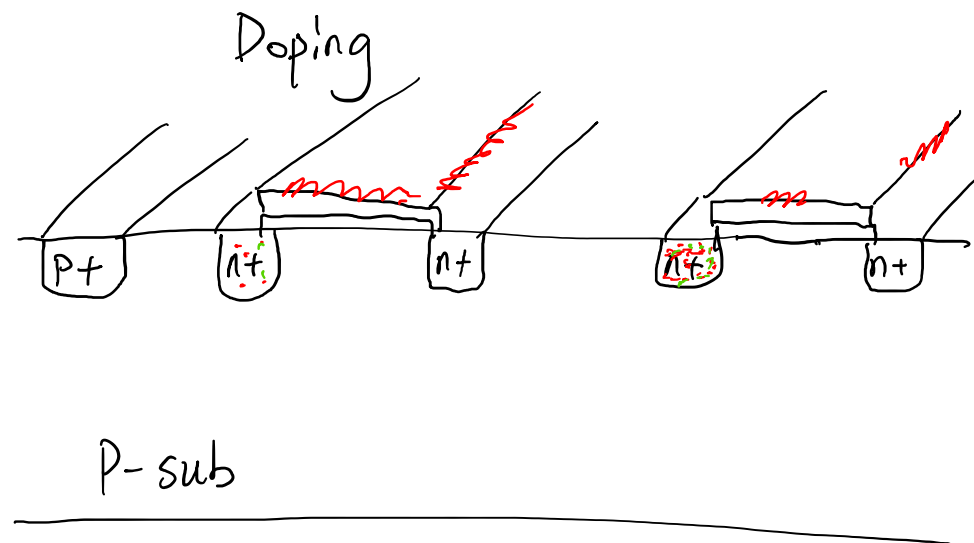
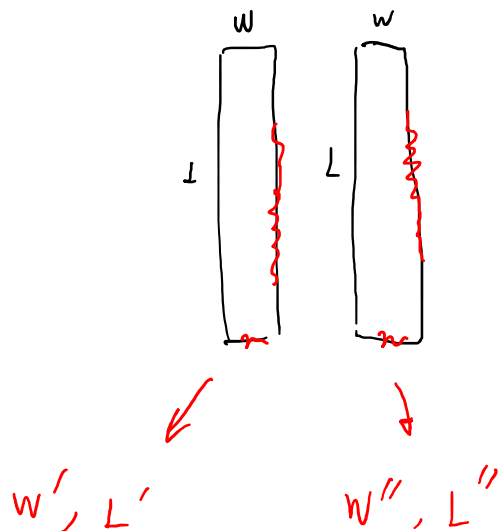
$$\Delta R = R_1 - R_2$$

Component Mismatch



Sources of Mismatch ...

Metal Roughness



Component Mismatch



Statistical (Empirical) Model

The mismatch between two identical devices,

$$\sigma^2(\Delta X) = \frac{A_X^2}{W \cdot L} + S_X^2 \cdot D^2$$

standard deviation

device parameter
(i.e., V_{th} , β , C , R ...)

Pelgrom
coefficient
(constant)

Size of
the devices

constant

physical distance between
the two devices.

Component Mismatch

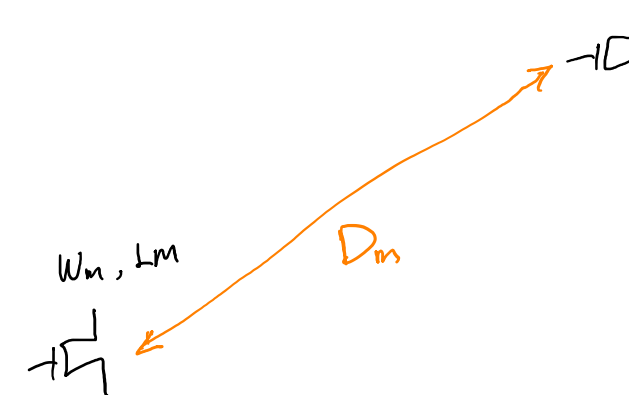


D_m is the distance between two identical devices for which the mismatch due to distance effect on parameter X is equal to the mismatch due to the size dependence for a minimum size device,

$$\text{i.e., } \begin{cases} W = W_m \\ L = L_m \end{cases}$$

$$\Rightarrow \frac{A_x^2}{W_m \cdot L_m} = S_x^2 D_m^2 \quad \Rightarrow$$

$$D_m = \frac{A_x}{S_x \sqrt{W_m \cdot L_m}}$$



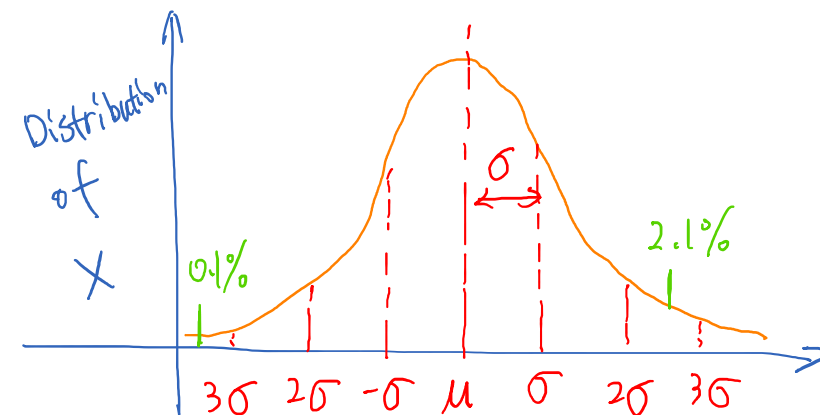
Typically, D_m is in the order of millimeters, hence we ignore the distance term.

Local Mismatch



⇒ "Pelgrom's Rule"

$$\sigma(\Delta X) = \frac{A_x}{\sqrt{W \cdot L}}$$



The device-to-device mismatch of parameter X for carefully laid out devices can be well-described by a Gaussian distribution with zero mean and standard deviation (σ) that depends on the process (A_x) and the size of the device (W and L).

Local Mismatch



$\sigma(\Delta X)$ measures the standard deviation of the difference of the parameter X of two devices.

For a random variable ΔX defined as $\Delta X = X_1 - X_2$, the variance

$$\sigma^2(\Delta X) = \sigma^2(X_1) + \sigma^2(X_2).$$

$$\Rightarrow = \sigma^2(X)$$

$$\Rightarrow \sigma^2(\Delta X) = 2\sigma^2(X)$$

$\Rightarrow$ For single device,

$$\sigma(X) = \frac{1}{\sqrt{2}} \sigma(\Delta X)$$

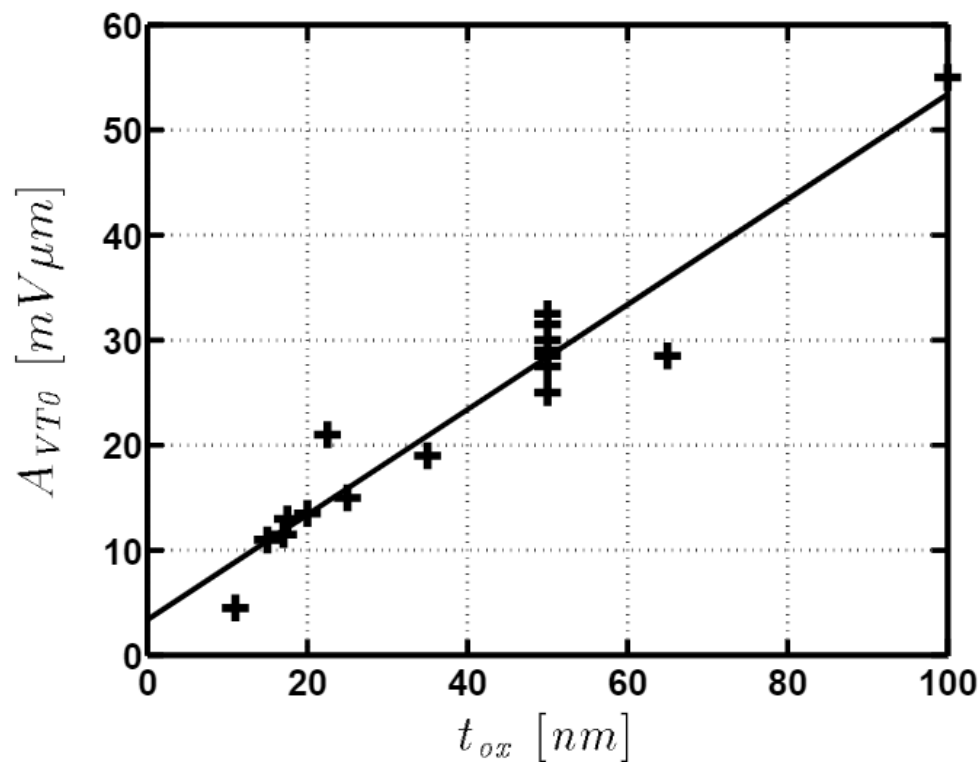
$$\text{eg., } \begin{cases} \sigma(V_{th}) = \frac{1}{\sqrt{2}} \sigma(\Delta V_{th}) \\ \sigma(\beta) = \frac{1}{\sqrt{2}} \sigma(\Delta \beta) \end{cases}$$

Local Mismatch

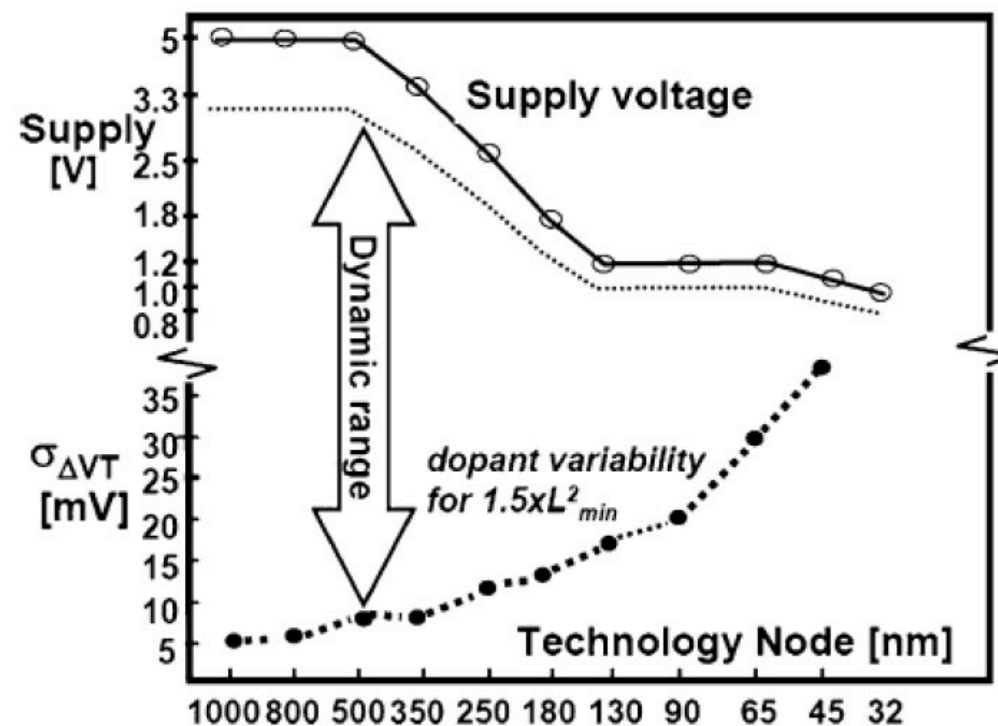


Mismatch vs. Scaling

Gate Thickness Dependency



Reduced Supply Voltage



Outline



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- Local mismatch
- Monte Carlo simulation

Monte-Carlo Simulation



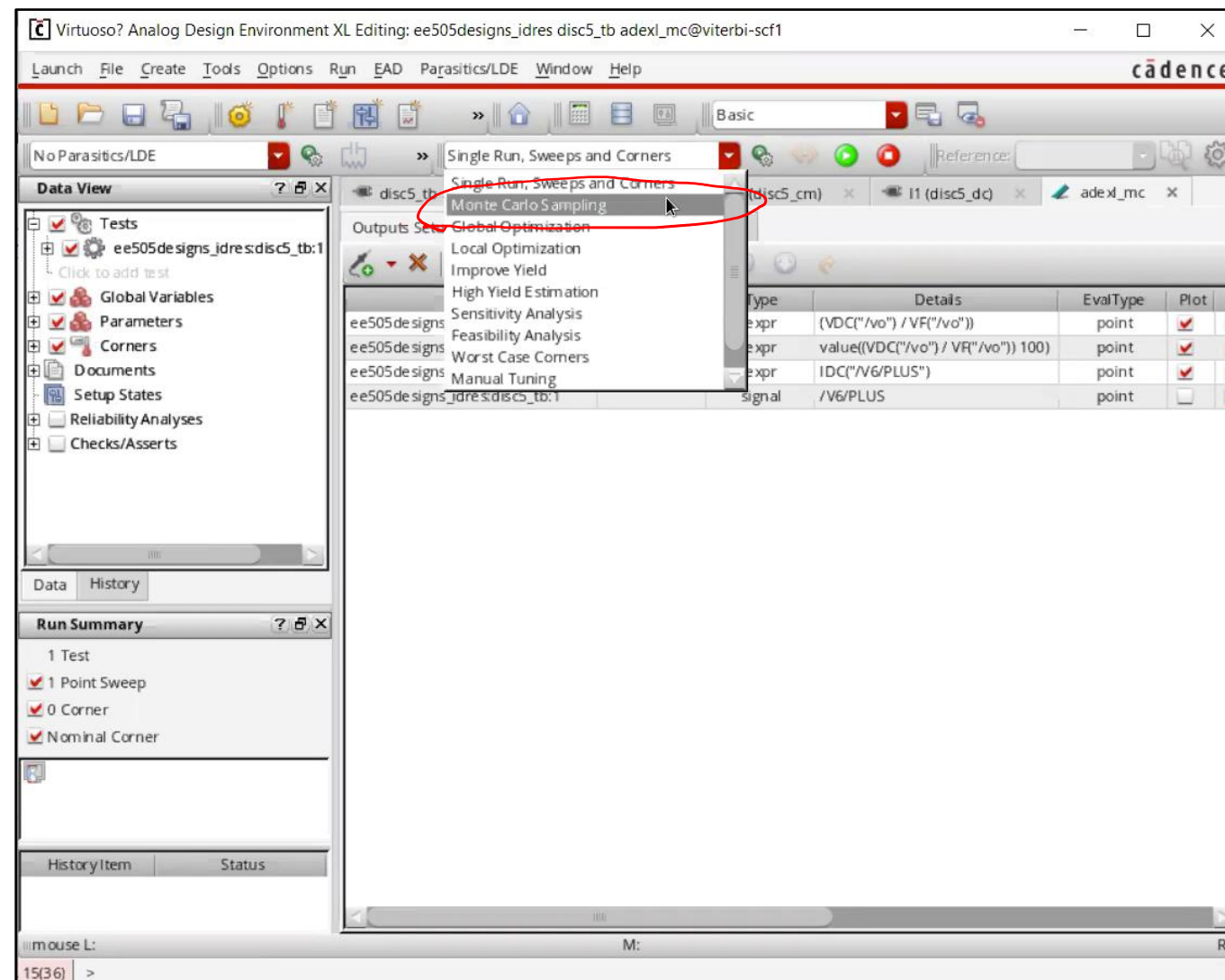
Monte-Carlo Simulation

— repeat random sampling to
obtain numerical results for
mismatch error estimation.

① ADE XL $\Rightarrow$

② ADE Explorer

★ The mismatch model assumes
near-ideal layouts of identical devices
with identical boundary conditions laid out
a few micron apart. ★



Monte-Carlo Simulation

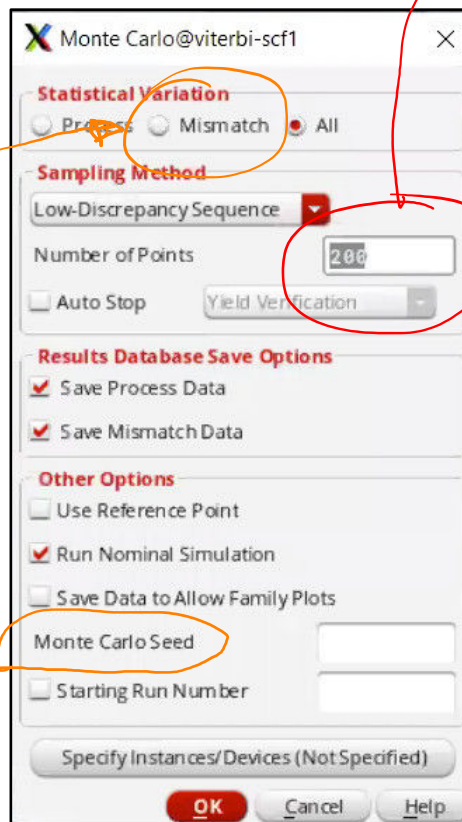


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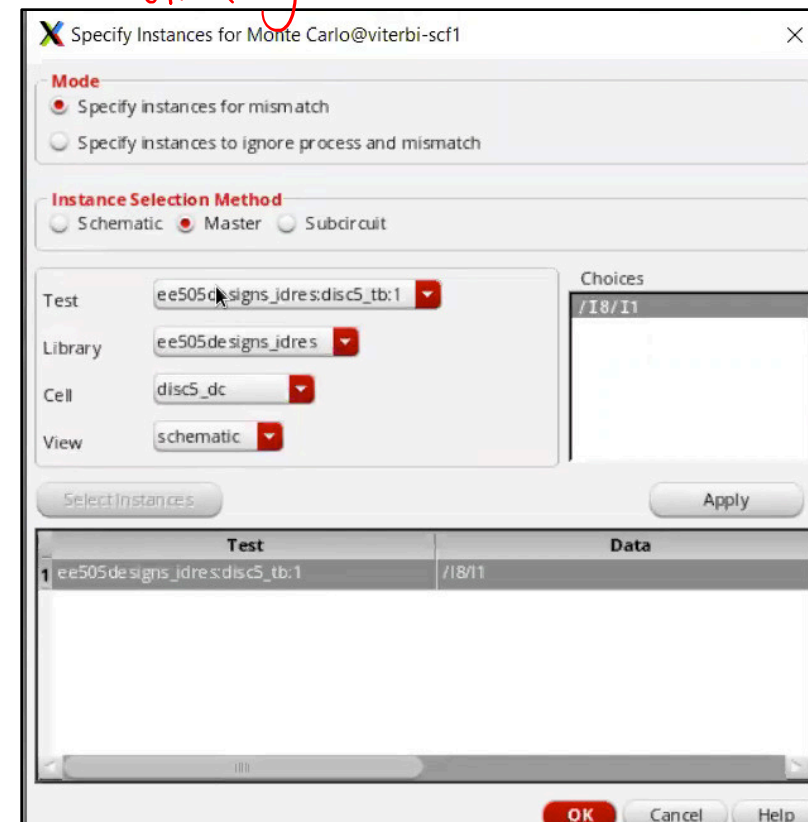
Process: Global or die-to-die variation

Mismatch: Intra-die variation
(EE 505) requires

same seed results in
same random samples
(For comparison purpose)



~100 will be sufficient, but keep increasing until you cannot see the changes.

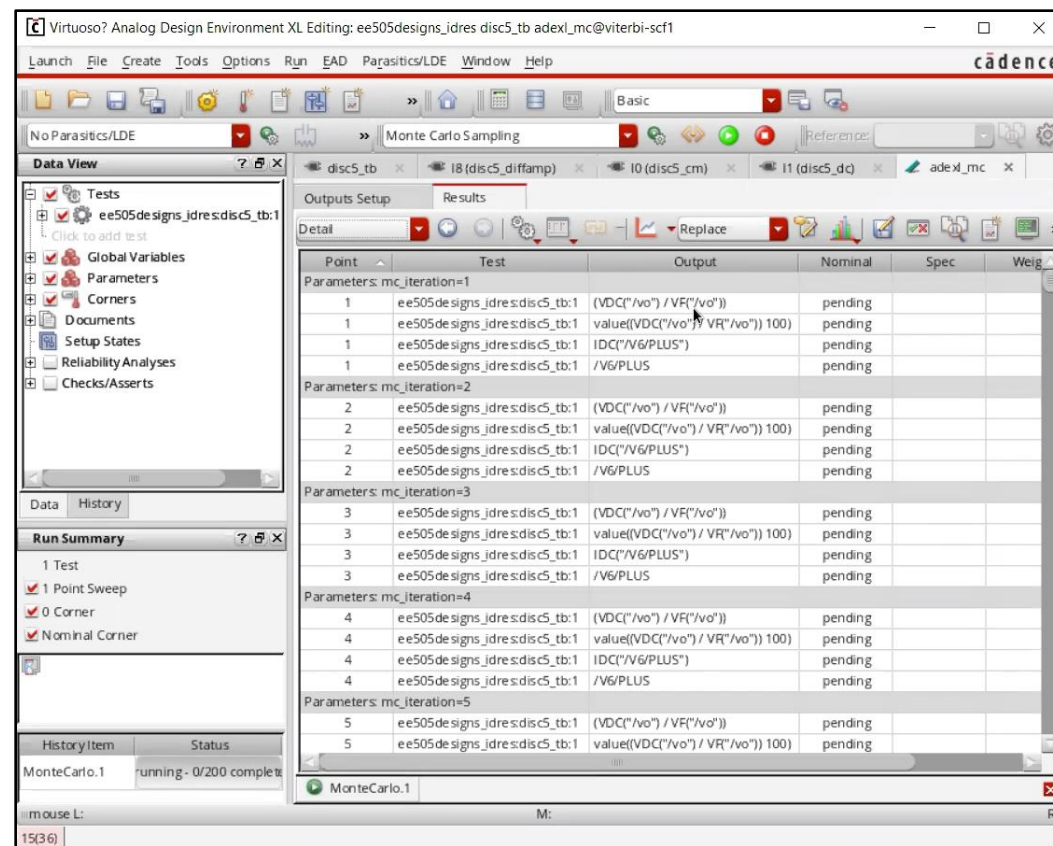
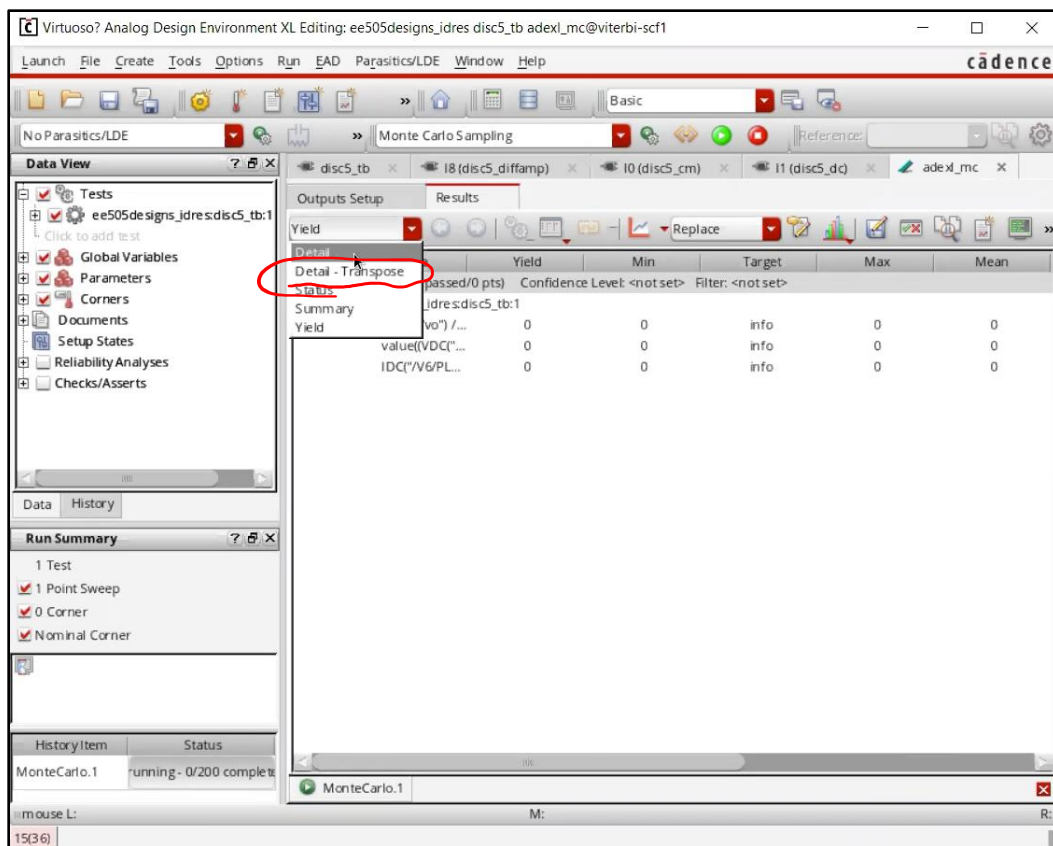


Monte-Carlo Simulation



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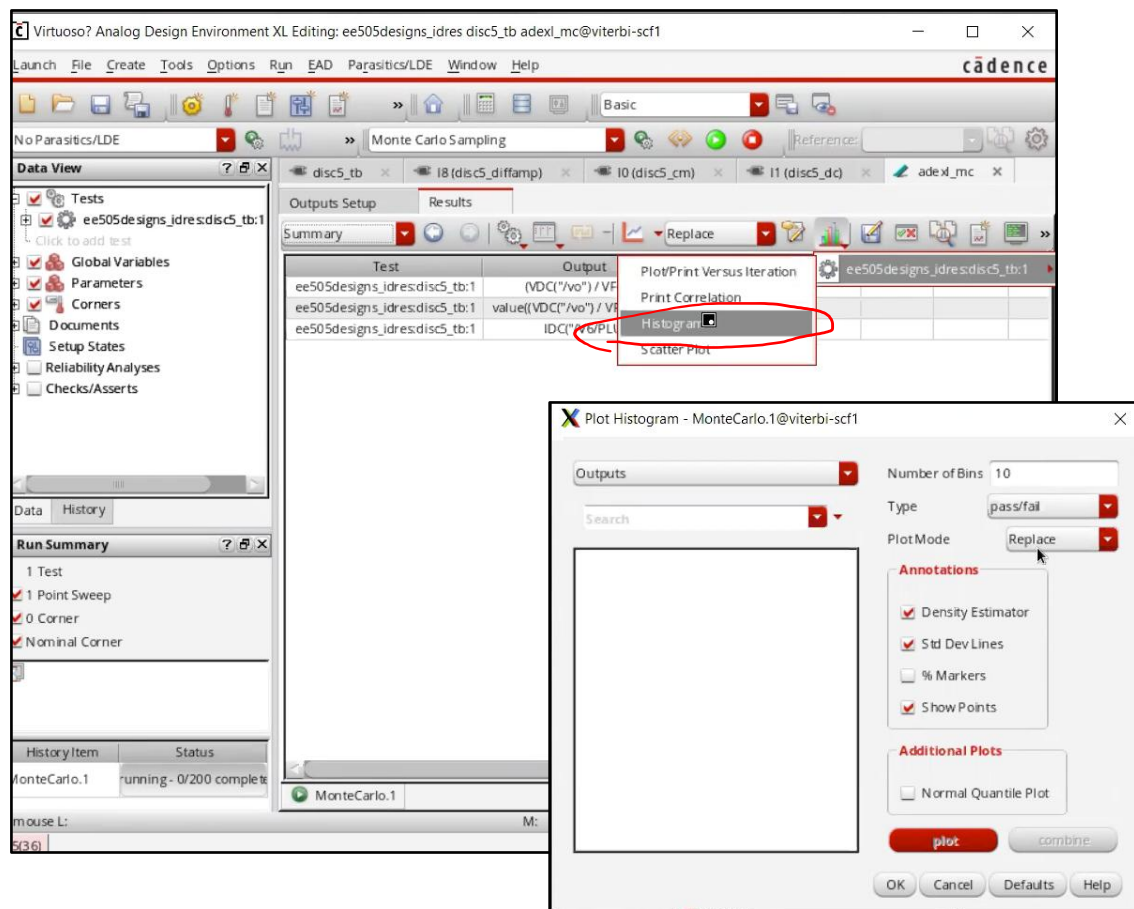
View the results



Monte-Carlo Simulation

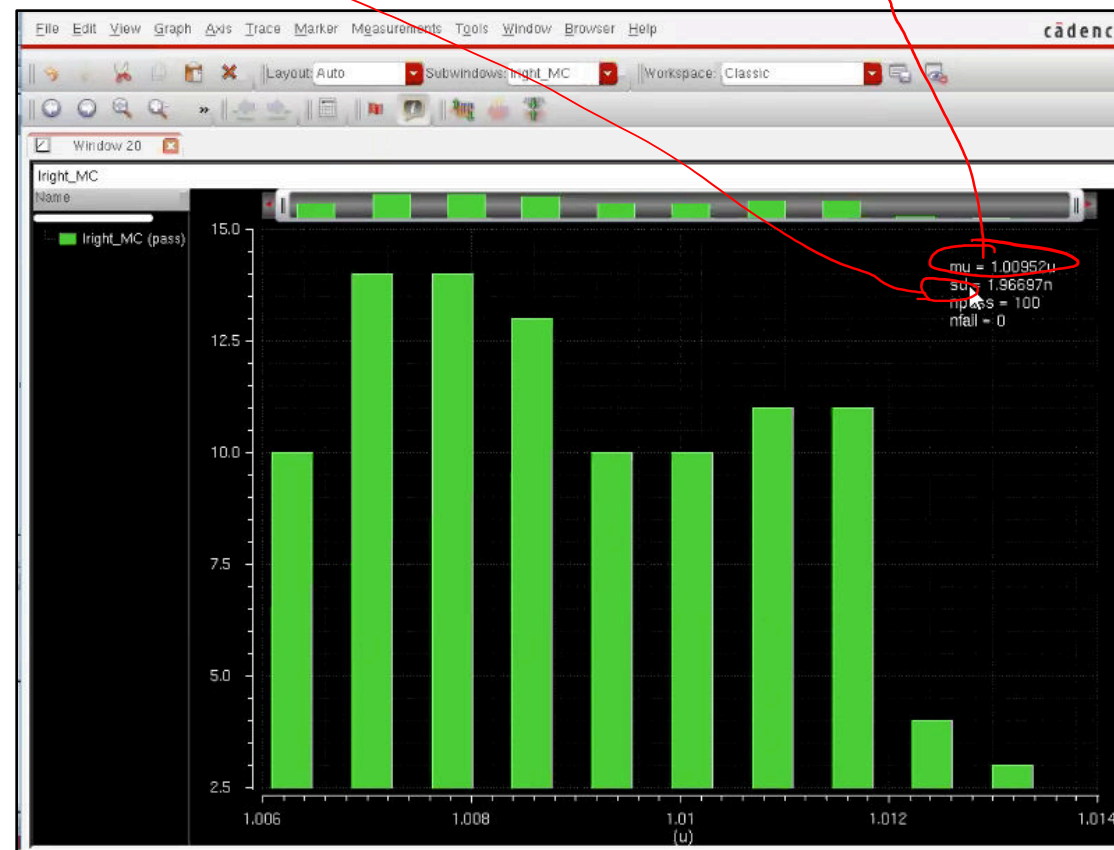


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standard deviation

mean value



Outline



- Analog layout basics
- Electro-magnetic interference (EMI)
- Pad and electro-static discharge (ESD)

Electro-Magnetic Interference (EMI)

EMI is the unwanted coupling of signals from one circuit/system to another.

⇒ through parasitic impedances (Capacitive, Inductive)

⇒ through supply/ground, bias, substrate (conductor)

⇒ through radiation (electro-magnetic)

↳ when inductor is used

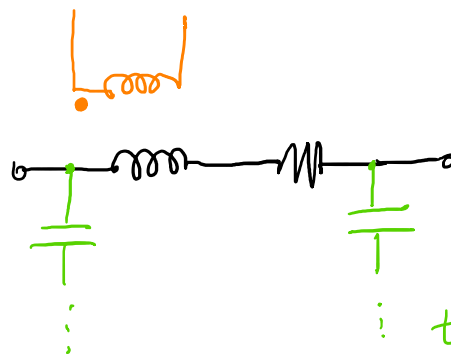
↳ when there is large return current loop.

Note

Traces (planes) are not perfect equi-potential surfaces.



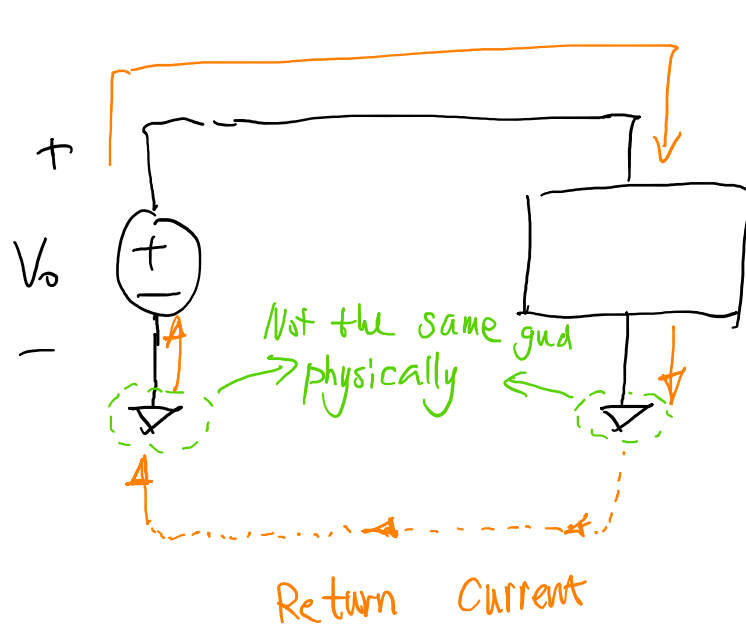
$\neq$



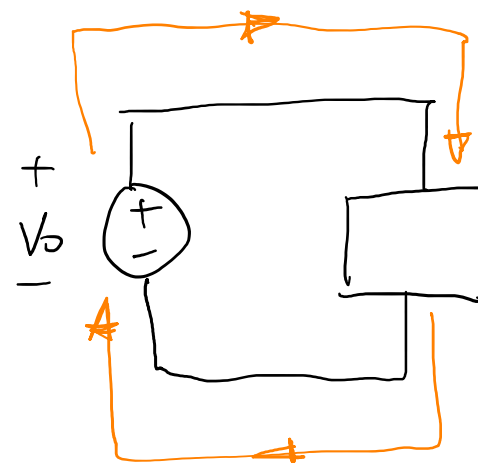
to gnd / other nodes.

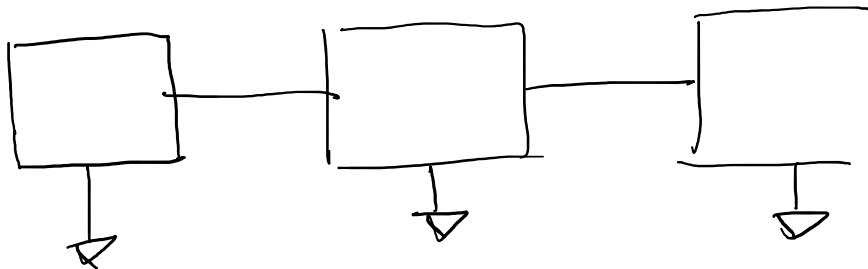
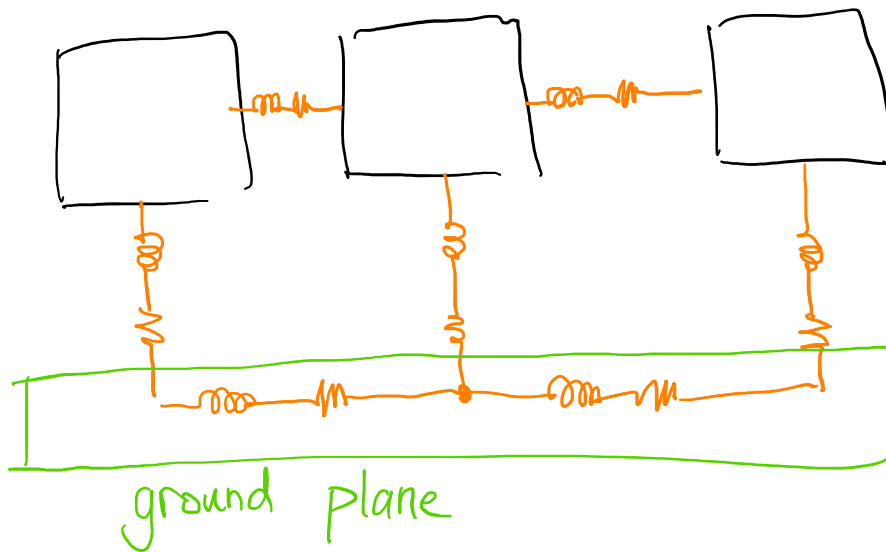
Note

Always follow the entire path of current.
(including the return path)



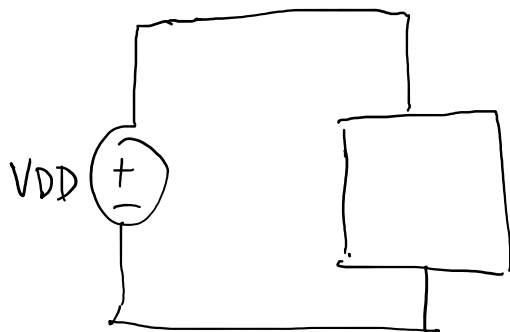
≡



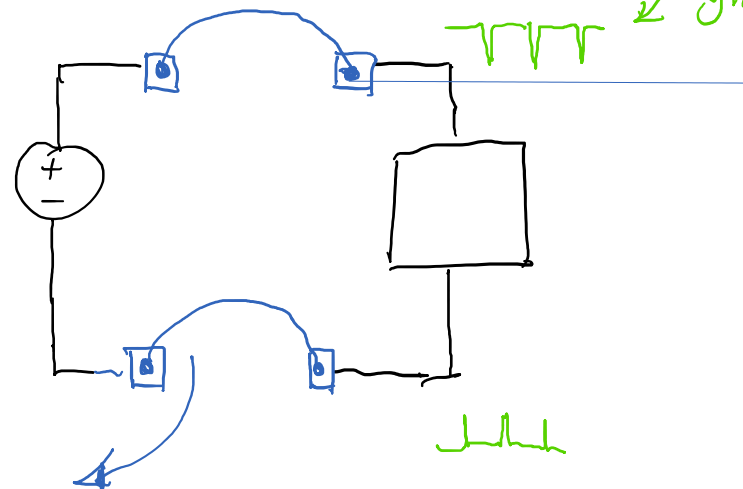
SchematicLayout

Supply Noise due to Switching Behavior

Ideal voltage source



In reality



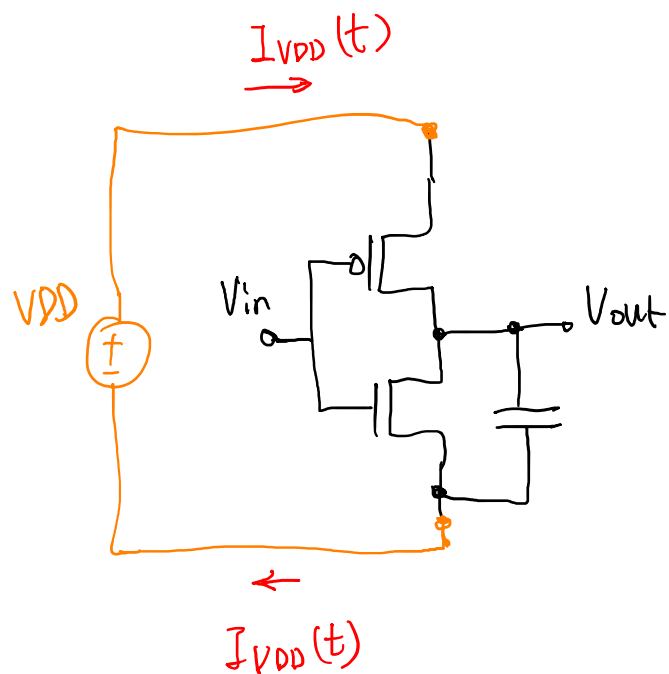
current or voltage
glitches.

1mm $\Rightarrow$ 1nH

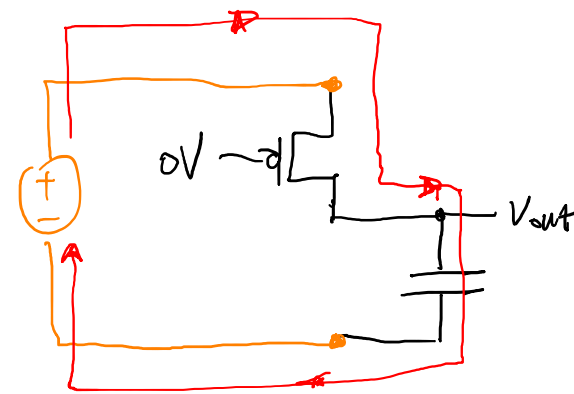
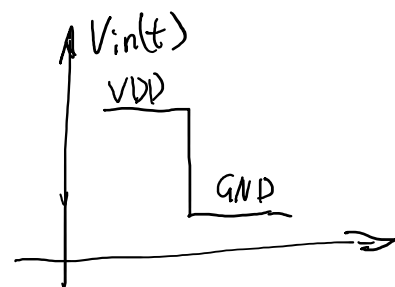
Bondwire
lumped Model



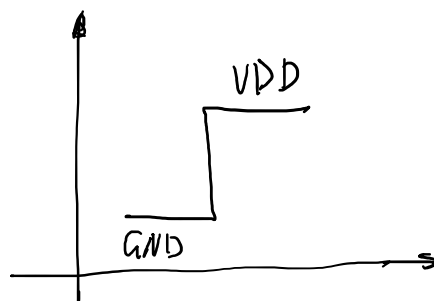
Example:



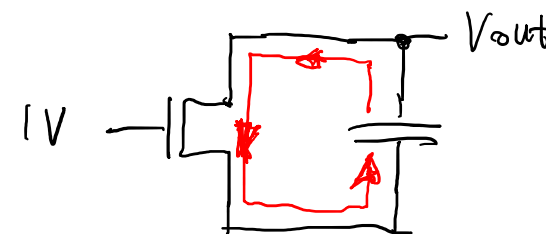
Case 1

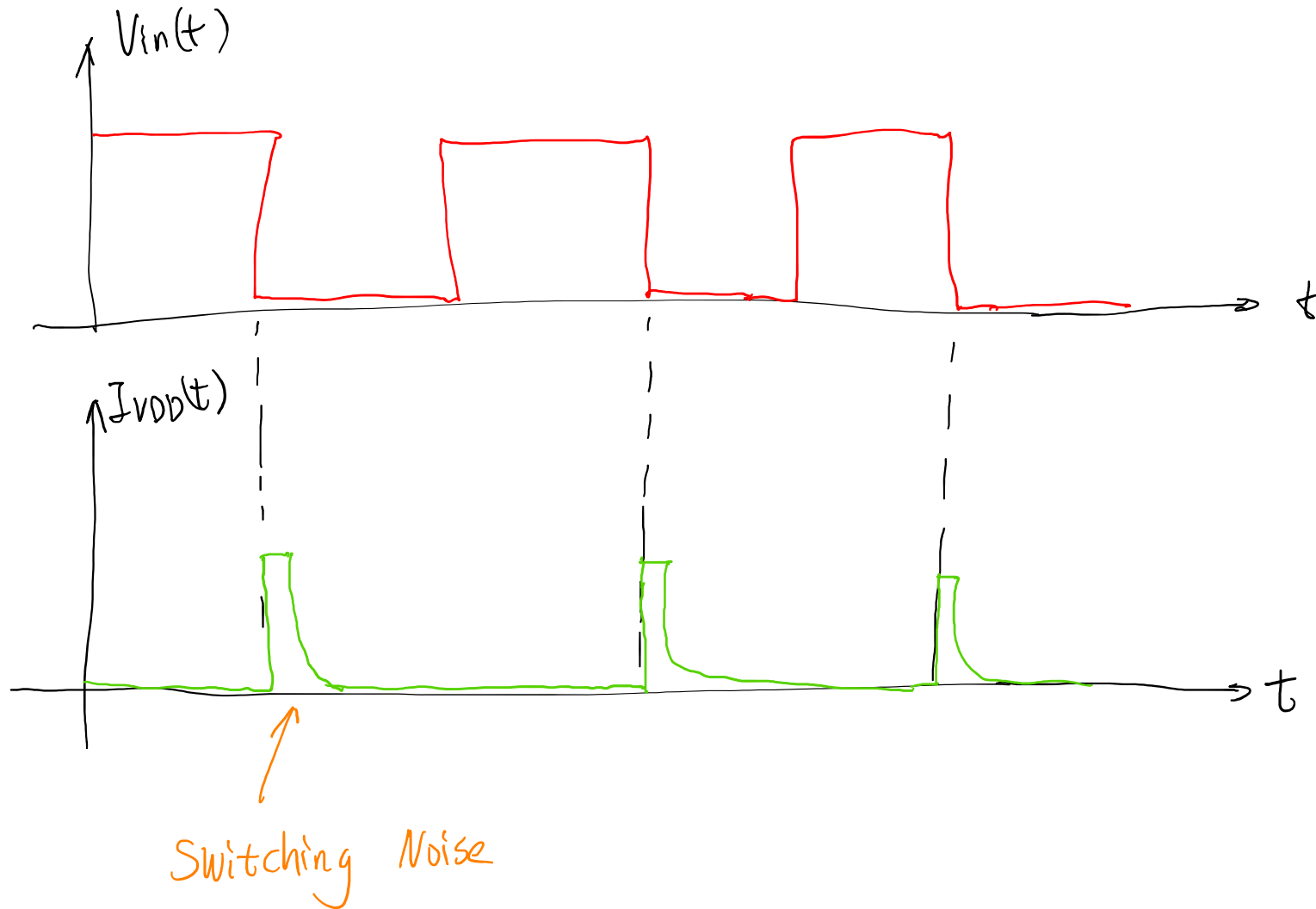


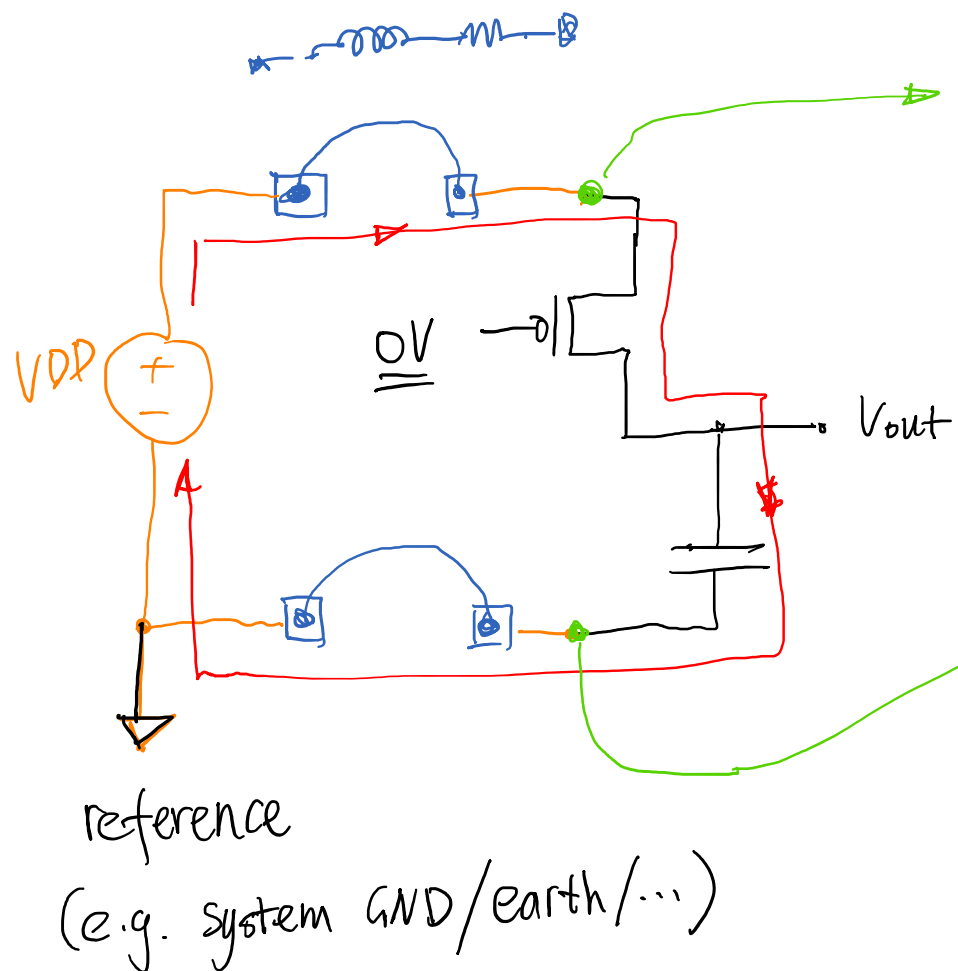
Case 2



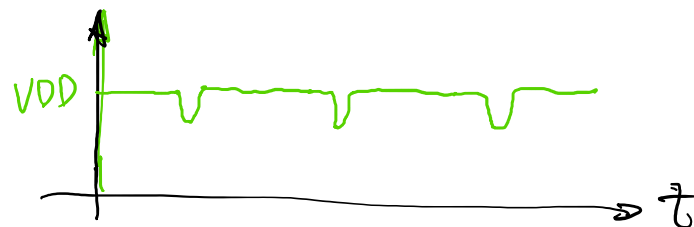
No current from the supply.



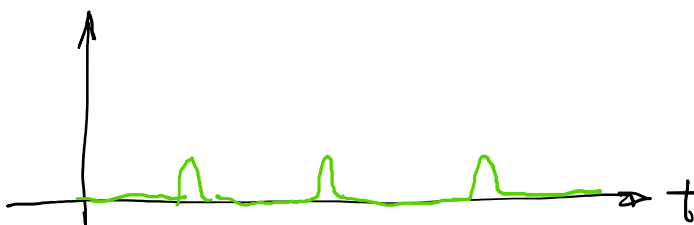




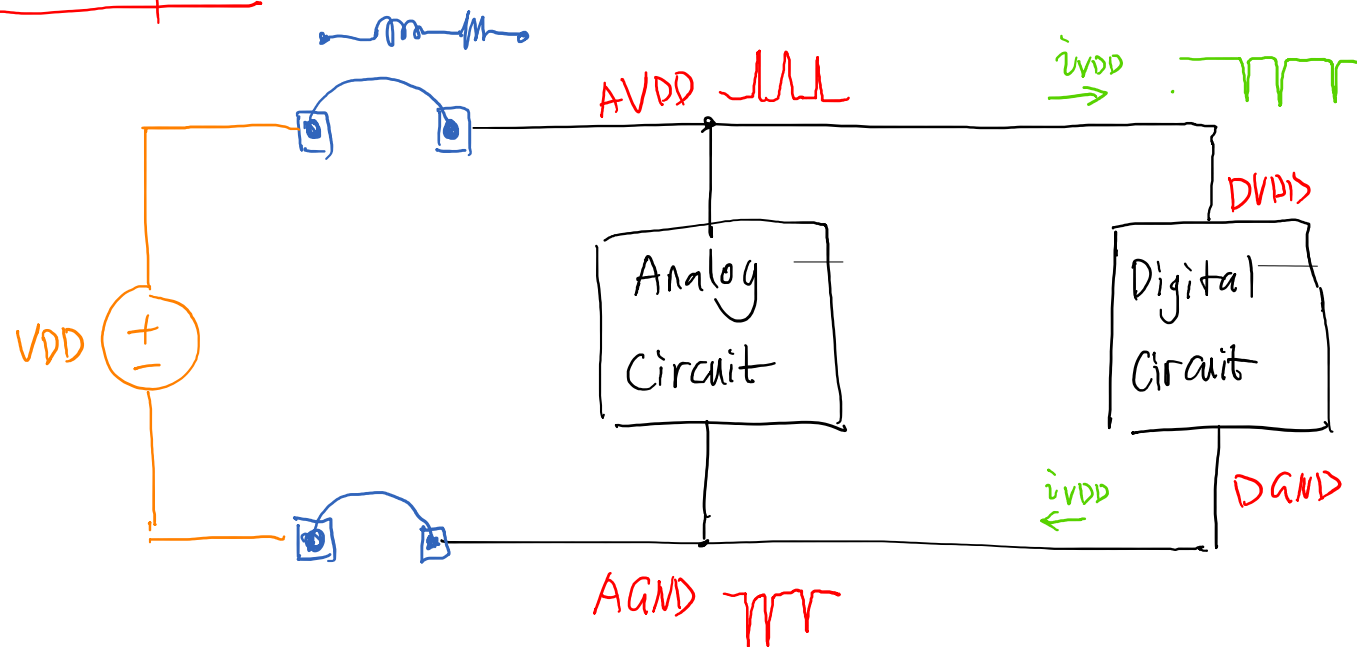
$$V_{DD} - L \frac{d}{dt} I_{VDD}(t) - R I_{VDD}(t)$$



$$L \frac{d}{dt} I_{VDD}(t) + R I_{VDD}(t)$$

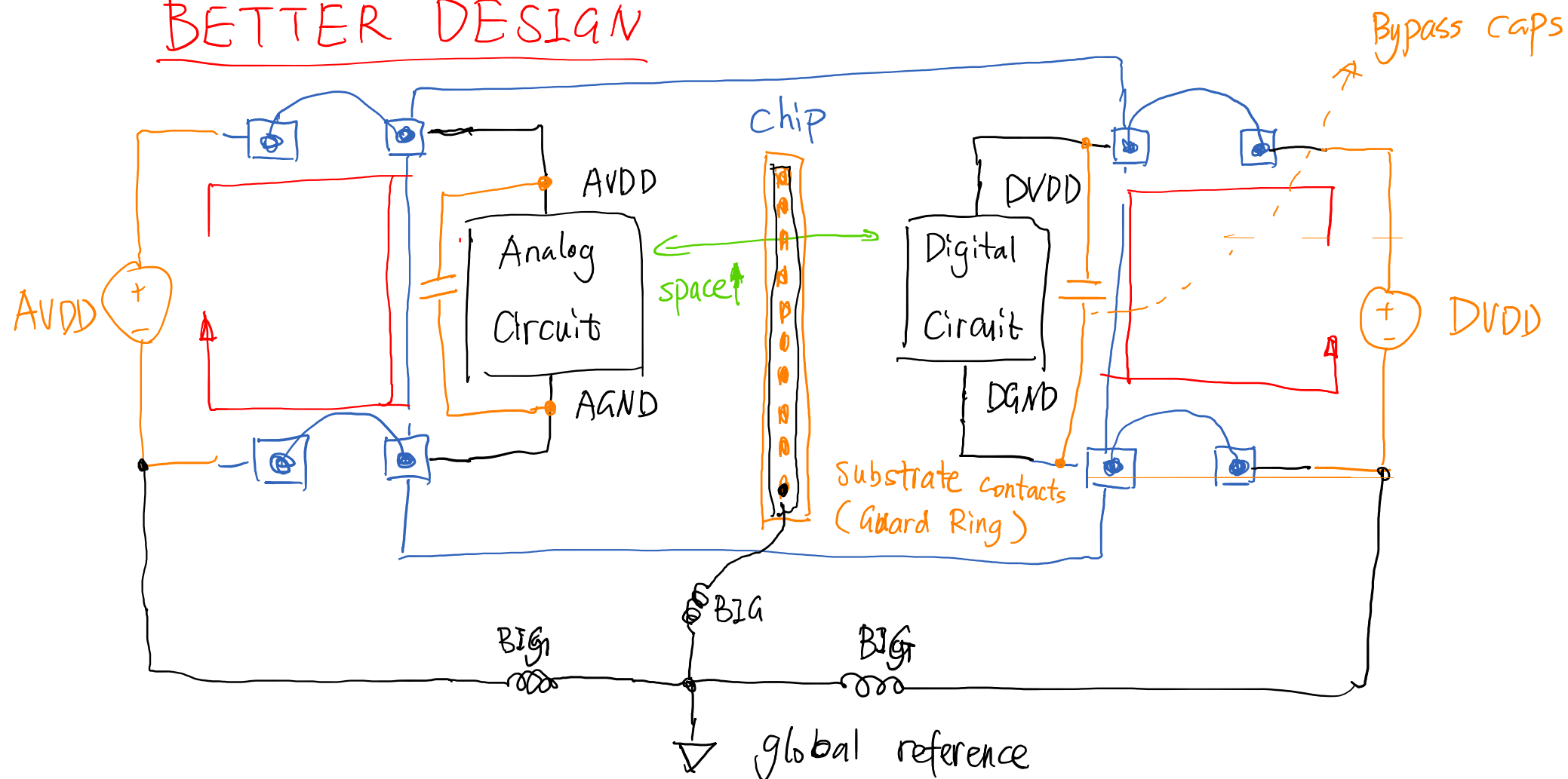


Example

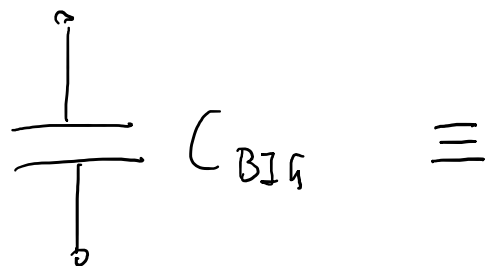


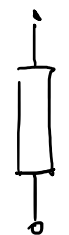
X BAD DESIGN

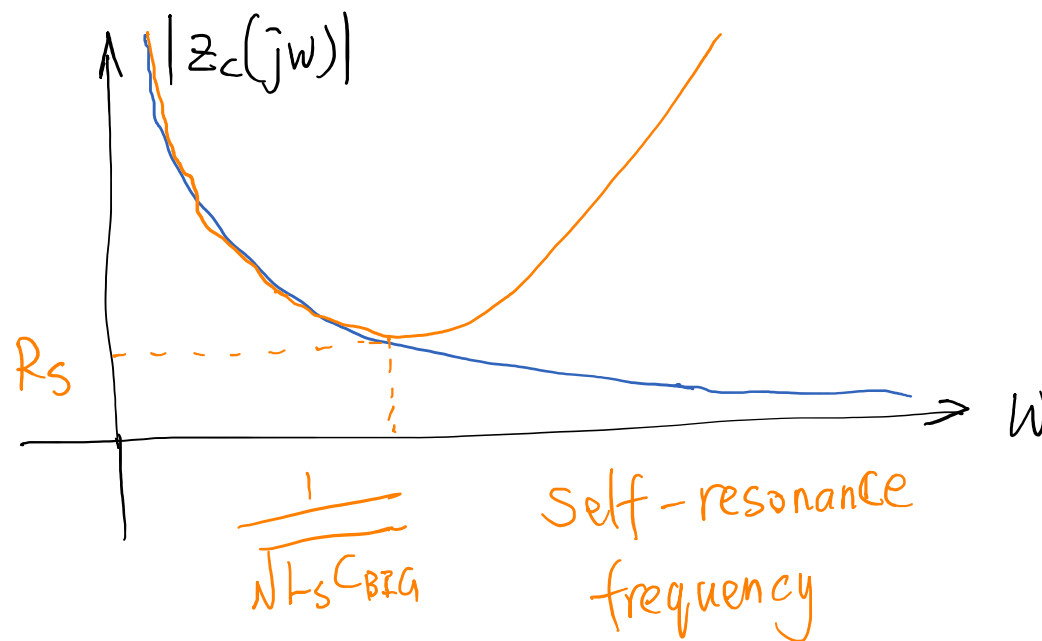
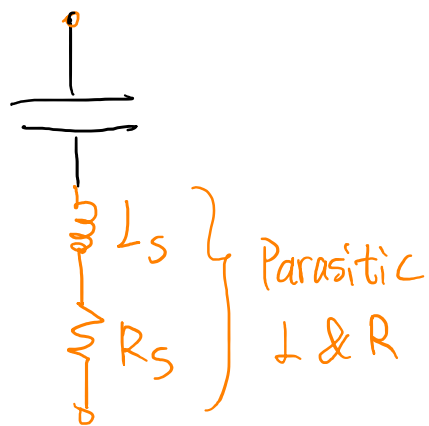
BETTER DESIGN

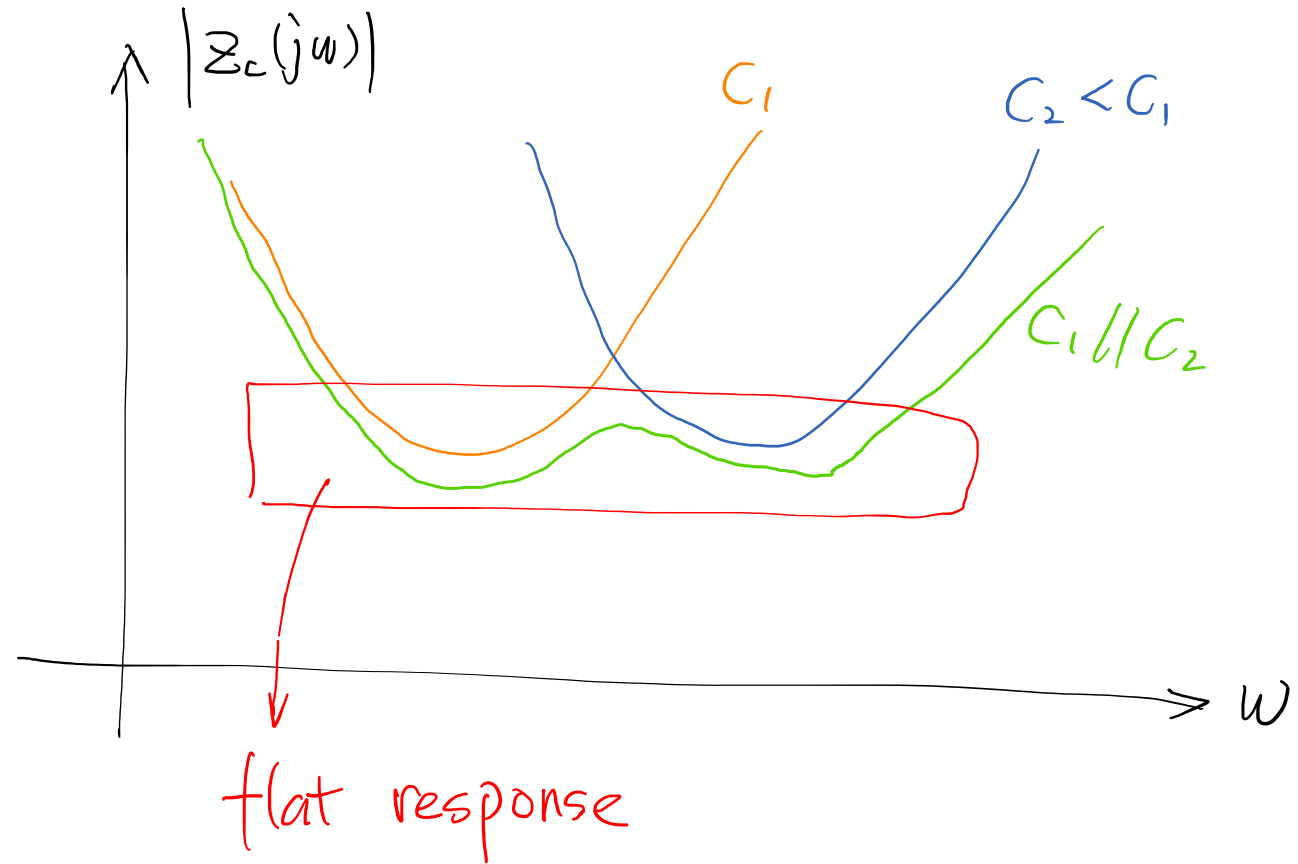
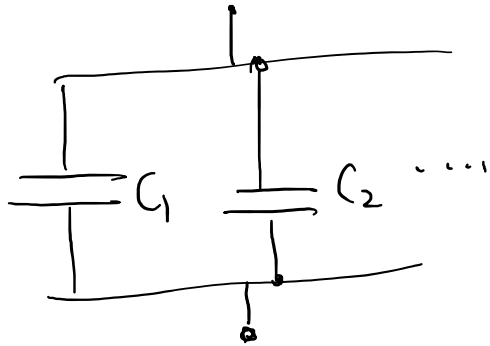


Bypass Capacitor

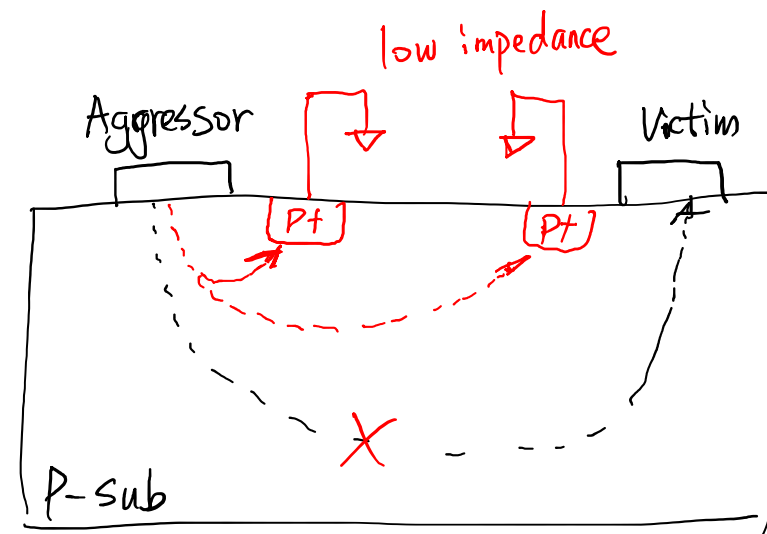
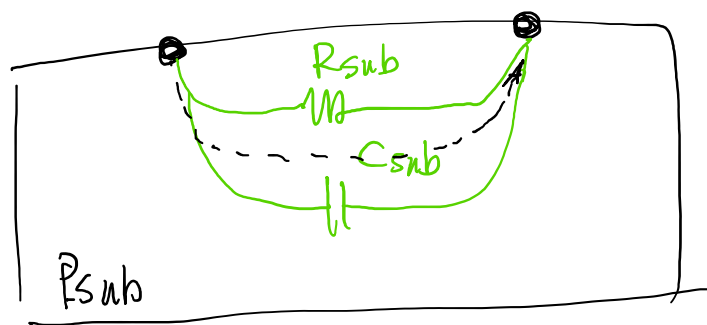



$$|Z_c(j\omega)| = \frac{1}{\omega C_{B1G}}$$



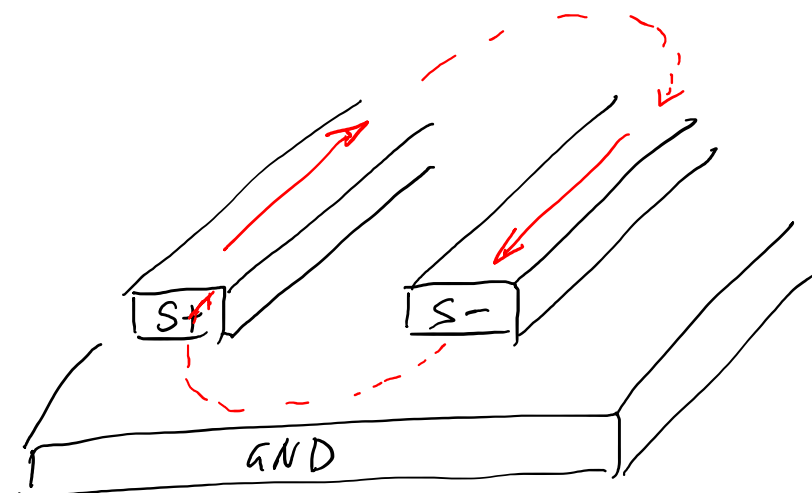
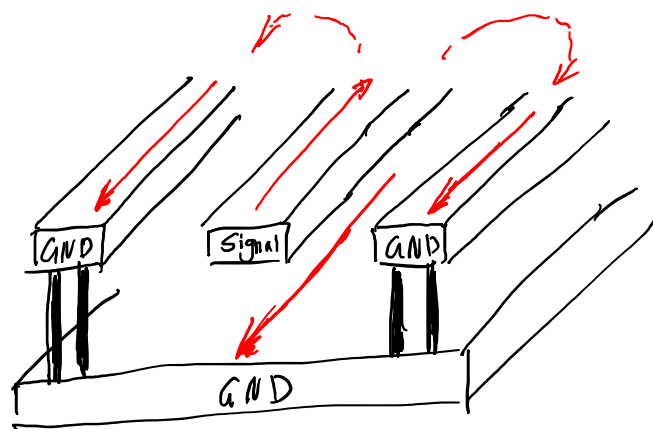
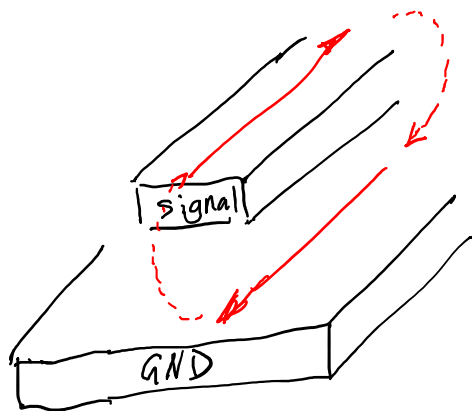
Solution

Substrate Coupling



★ Use substrate contacts
(Guard Ring)

Use Differential Signalling



✓✓ Less EMI

Outline



- Analog layout basics
- Electro-magnetic interference (EMI)
- Pad and electro-static discharge (ESD)

Pad



Pad



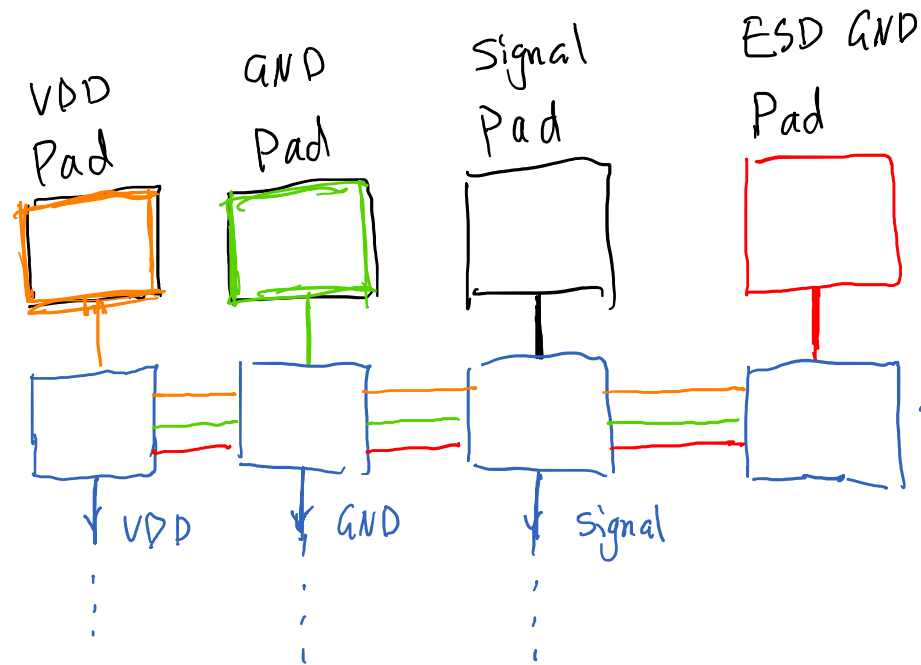
Side View



Pad

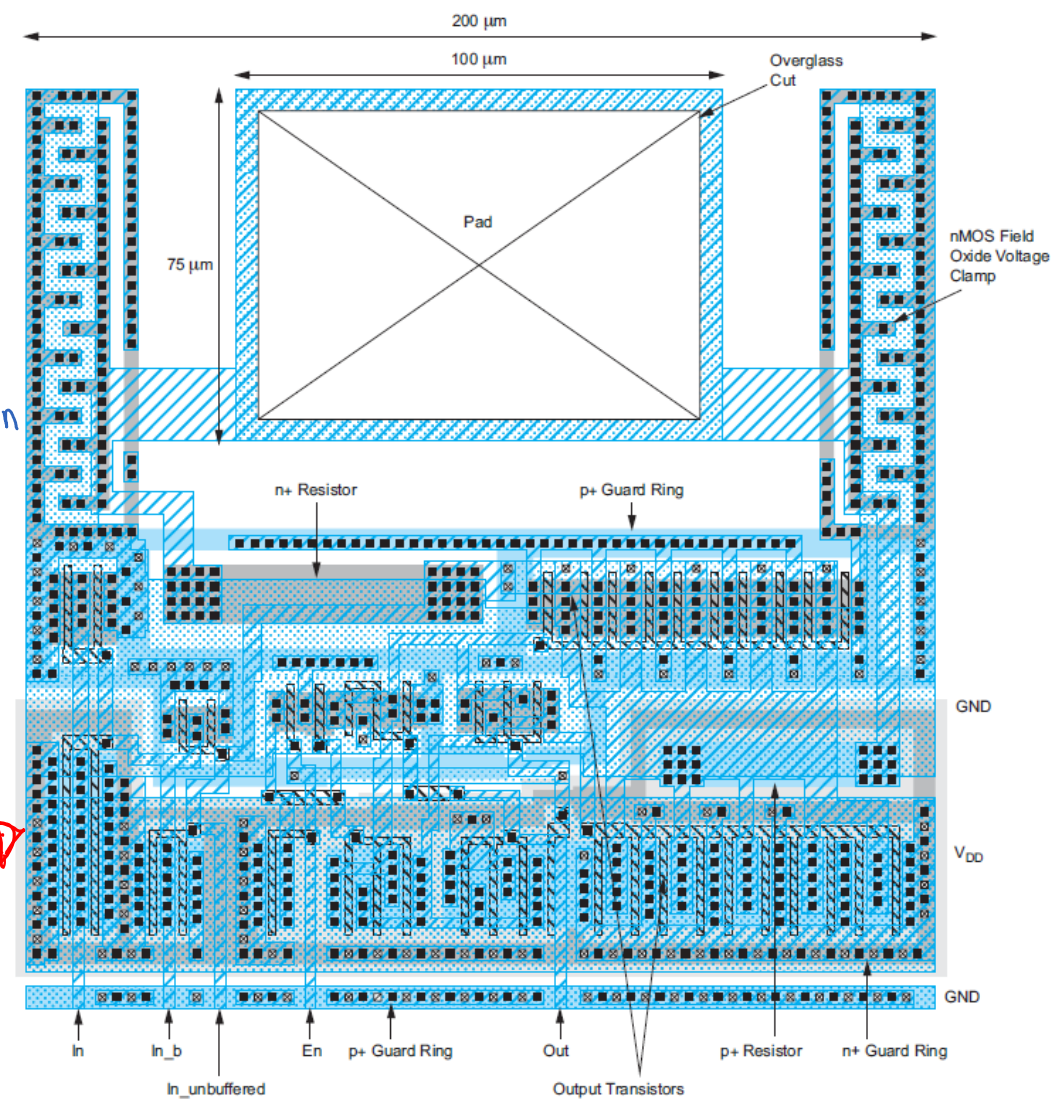


Pad Structure

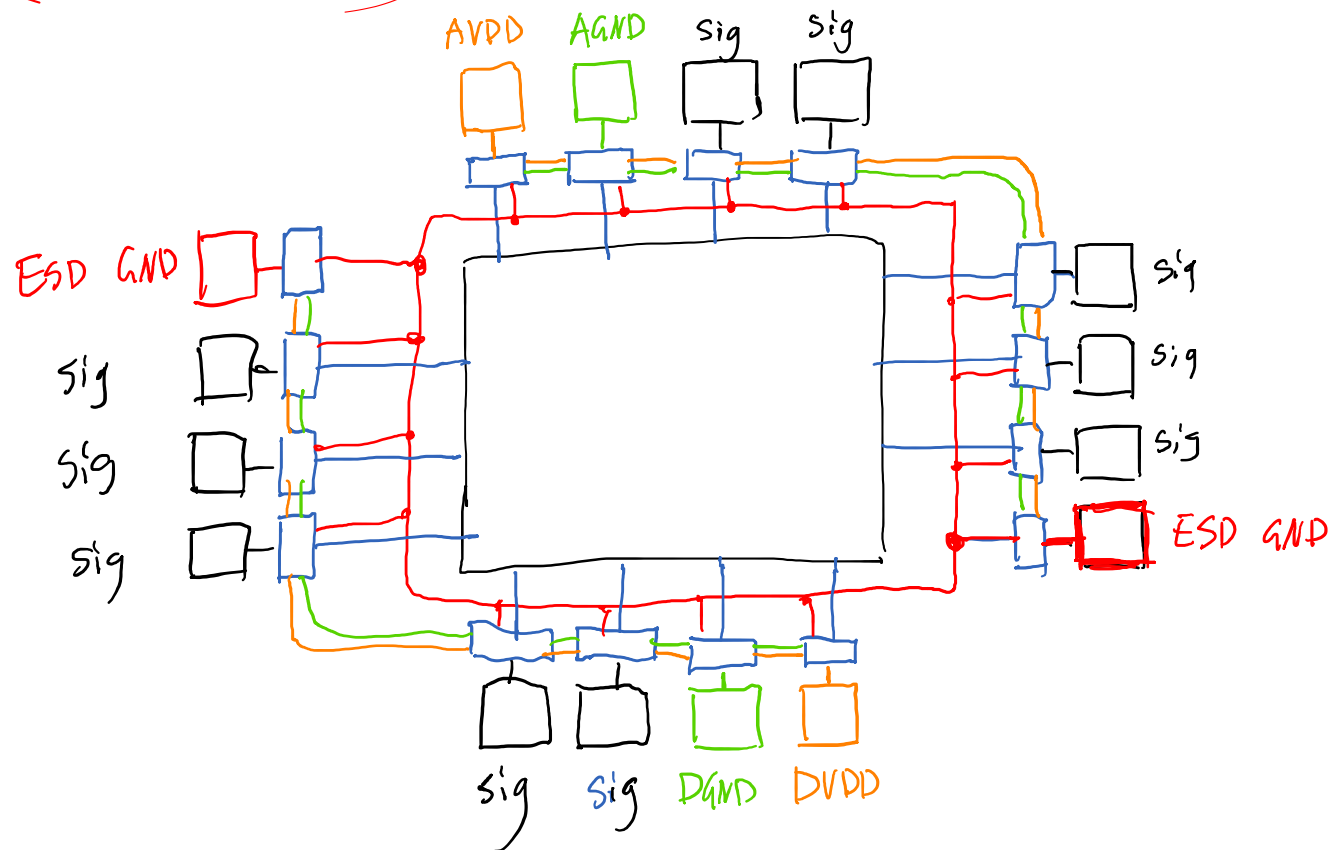


To the circuits

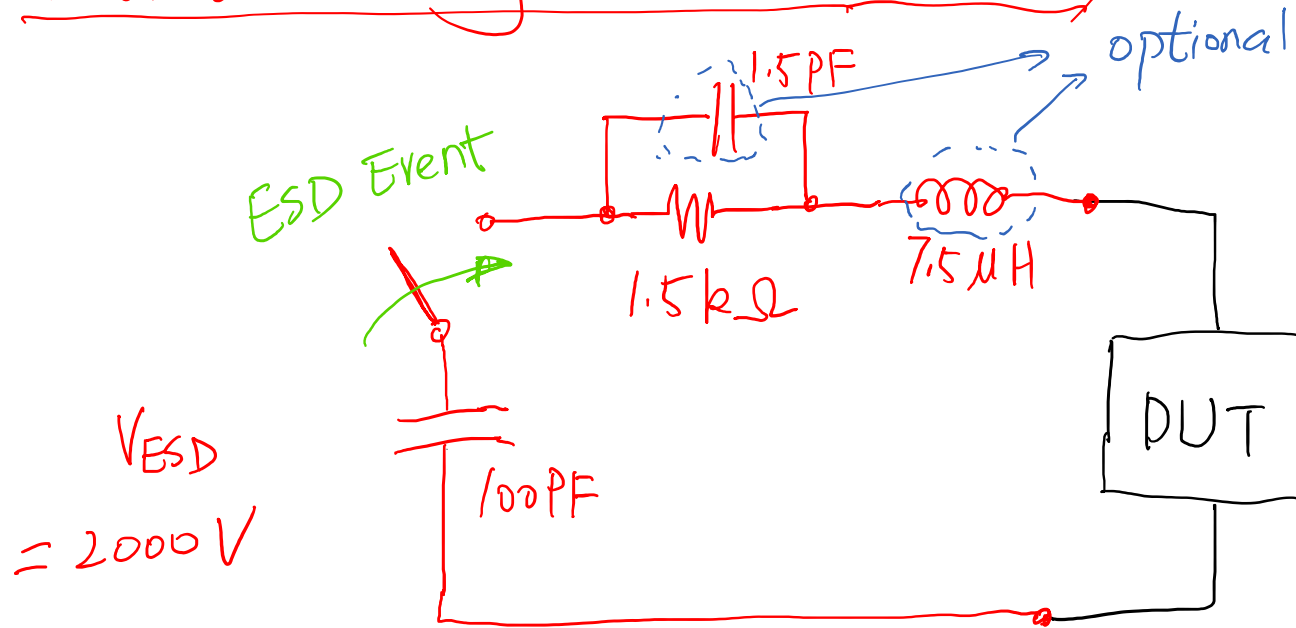
Signal pad example →



Example: AMS @chip



Human Body Model (HBM)

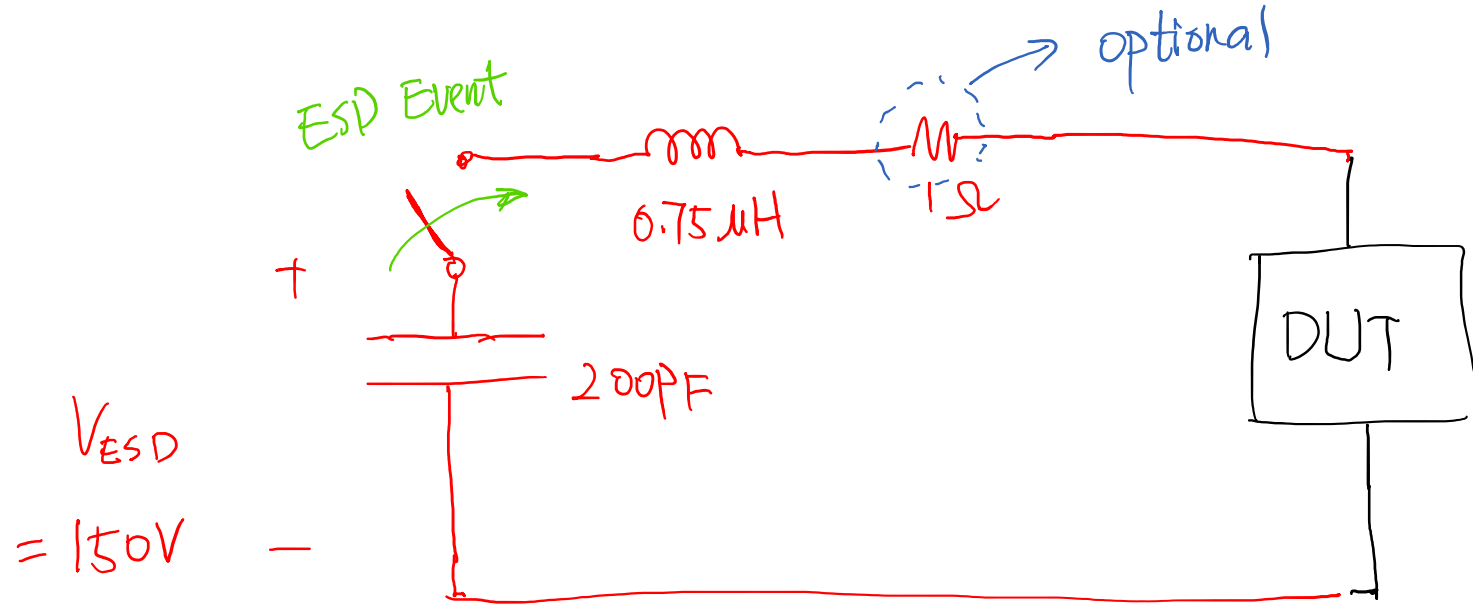


Device Under
Test

(Your design + ESD protection
Circuits)

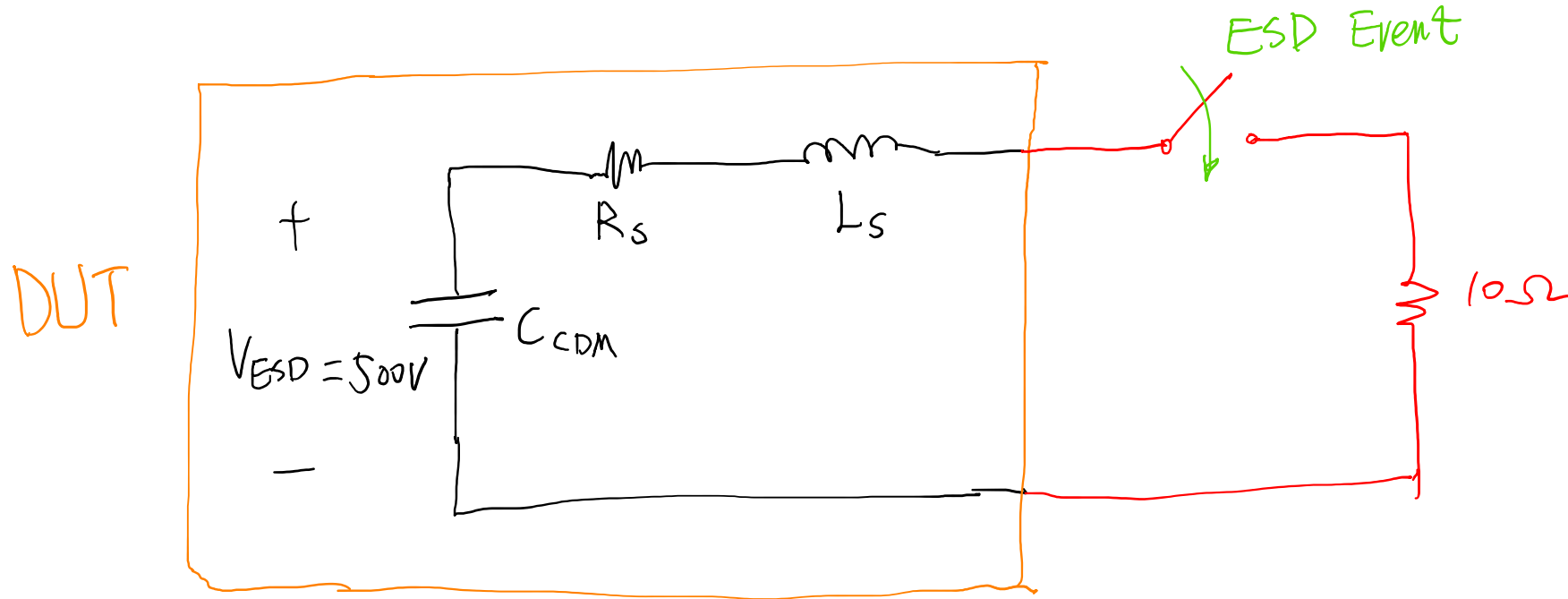
* Human body is charged, then discharged to DUT.

Machine Model (MM)



☆ Test equipment or PCB is charged, then discharged to DUT.

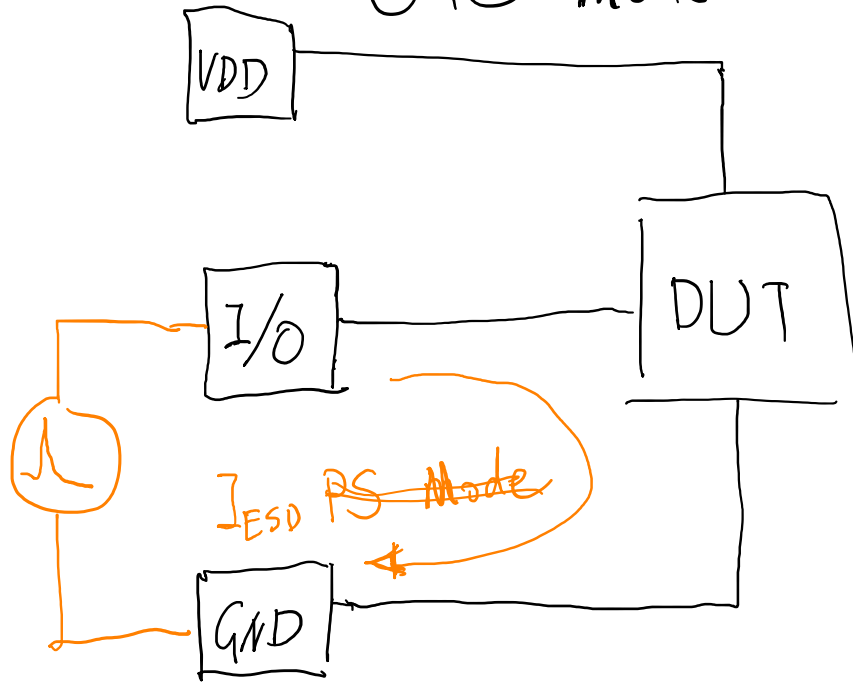
Charged Device Model (CDM)



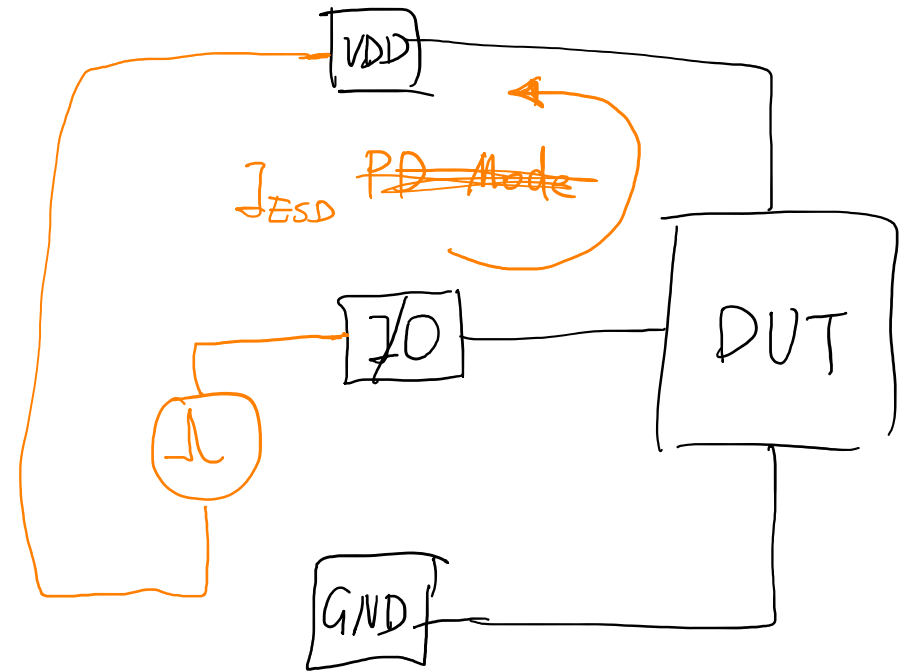
★ DUT is charged, then discharged to ground.

ESD Zapping Modes

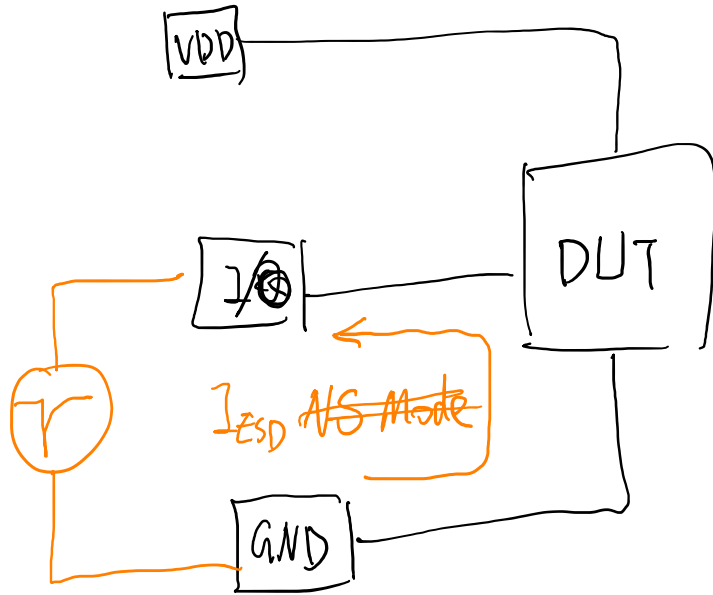
① PS Mode



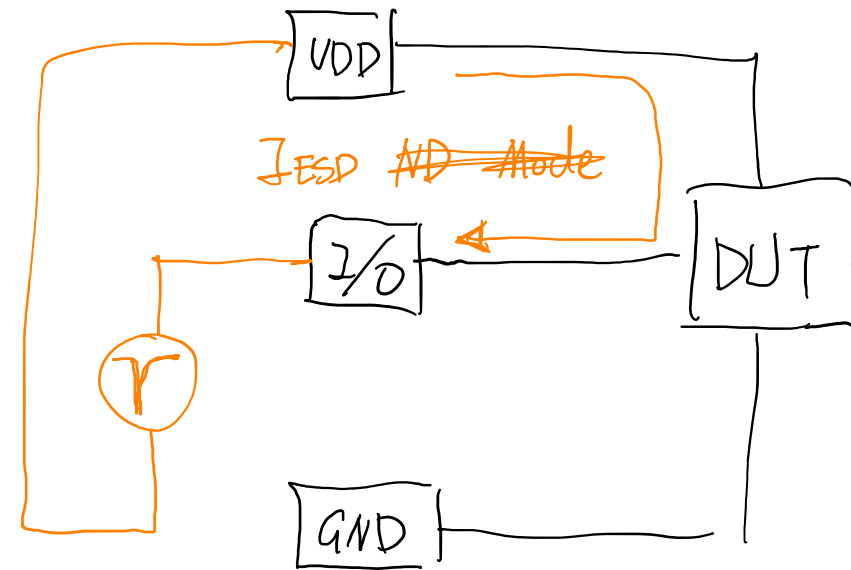
② PD Mode



③ NS Mode

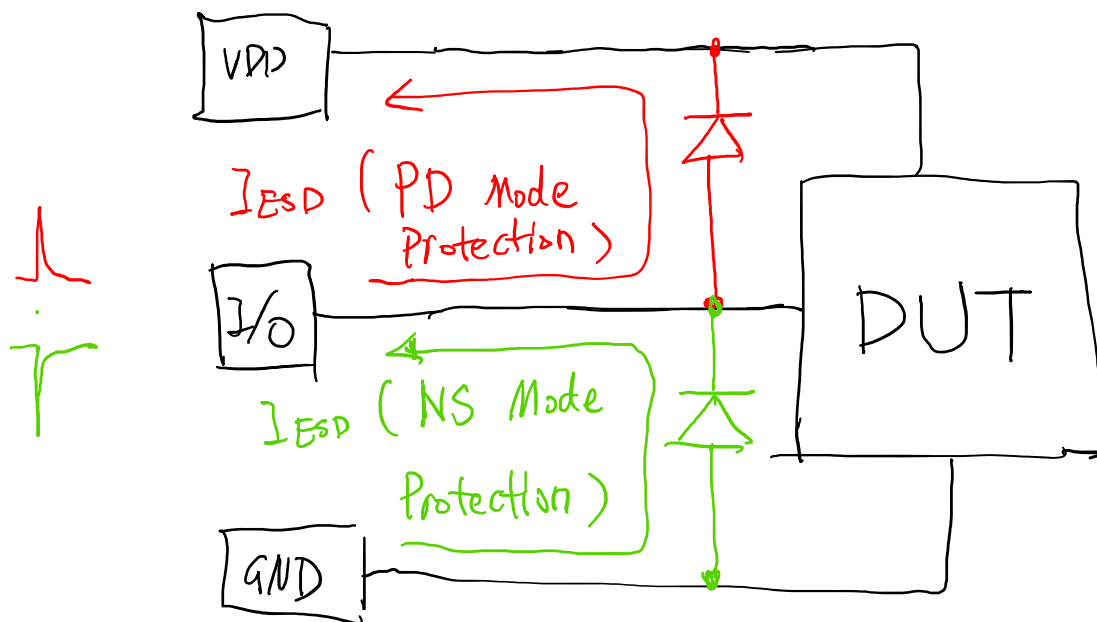


④ ND Mode

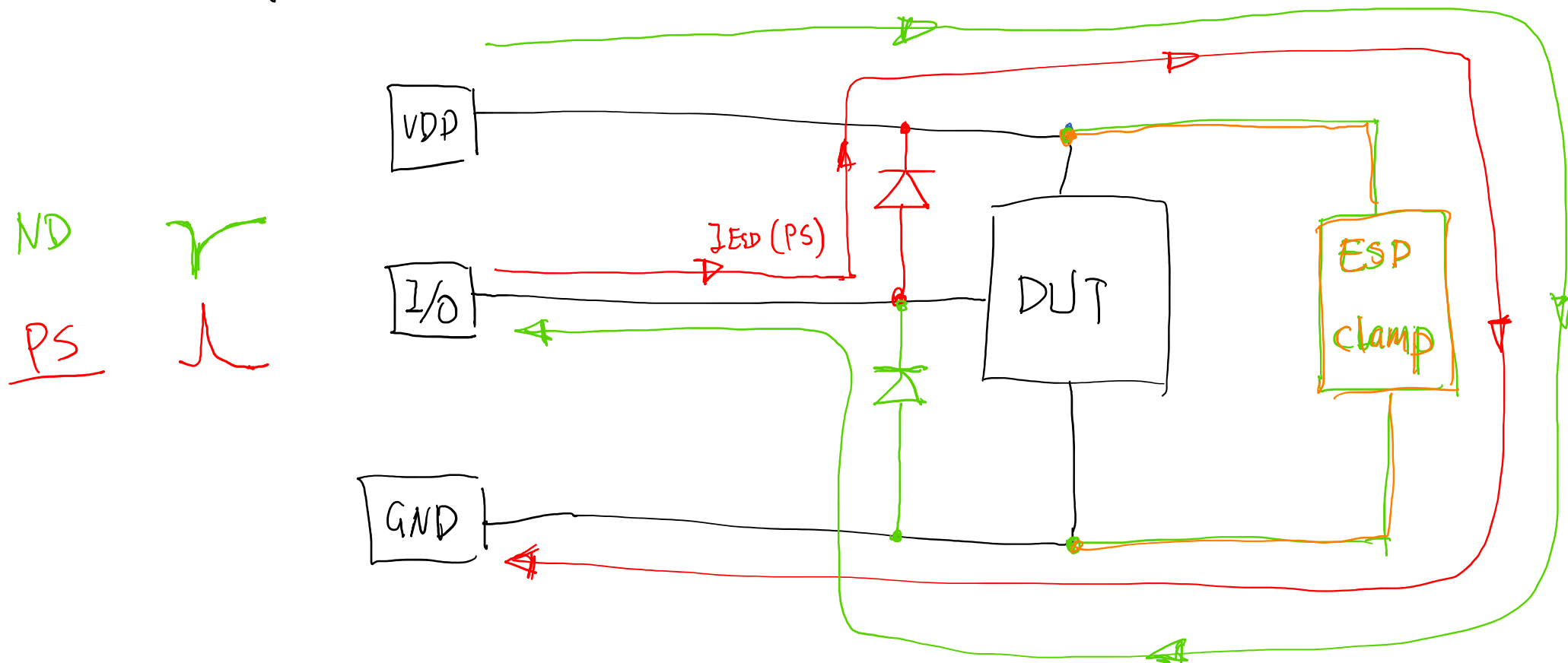


Non-Snapback-Based ESD Protection (Diode-Like)

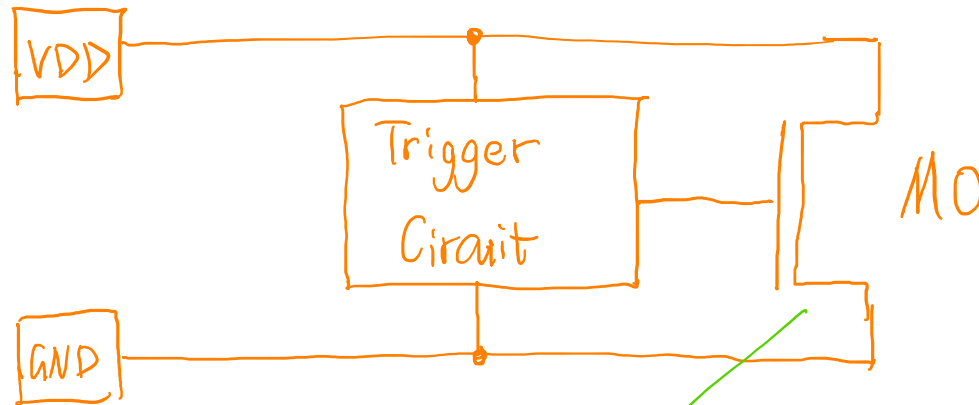
— Protect PD & NS mode stress. (diodes)



- Protect ND & PS mode stress. (diodes + clamp circuits)



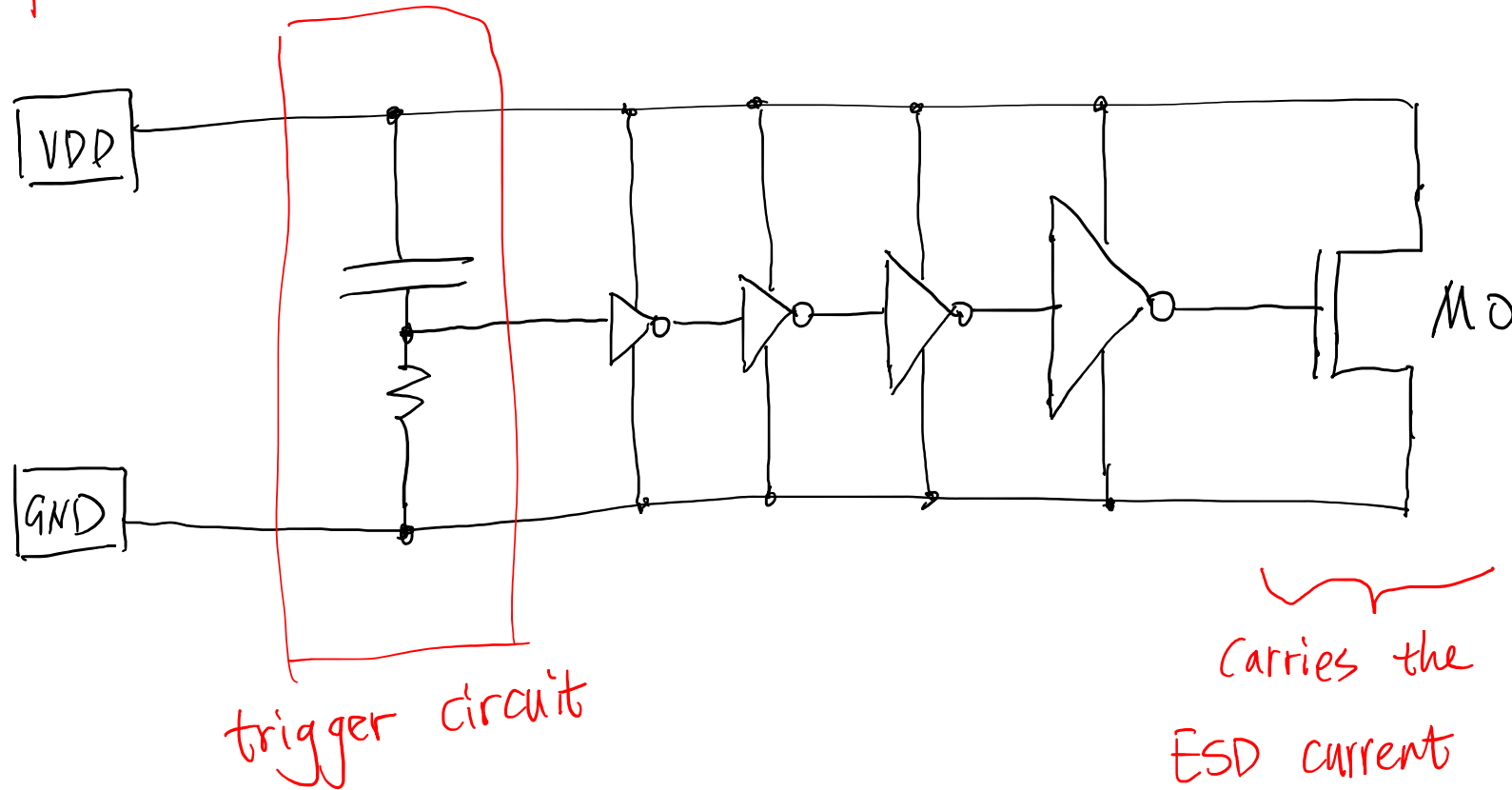
ESD Clamp Circuit



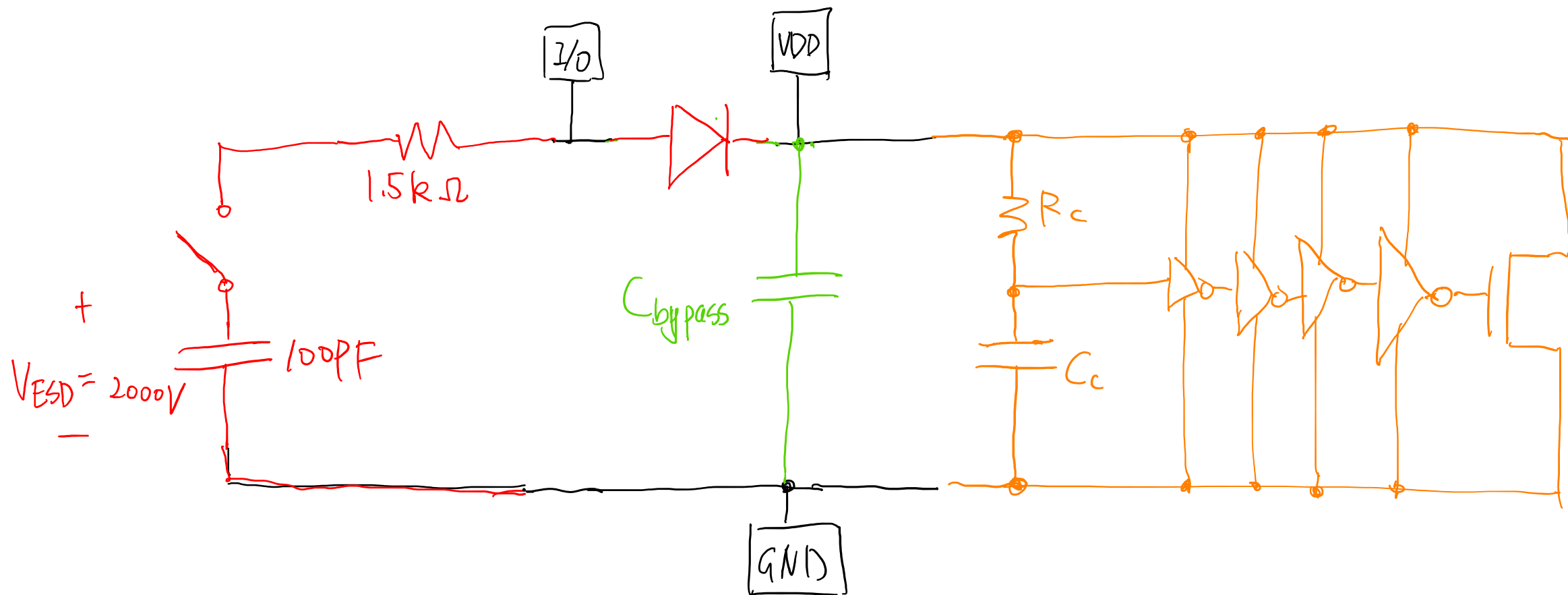
Big enough to carry
all the ESD current!

- ① Use trigger circuit to detect the ESD event.
- ② Turn on MO
- ③ Turn off MO after ESD event.
- ④ Do not turn on during regular power-up event.

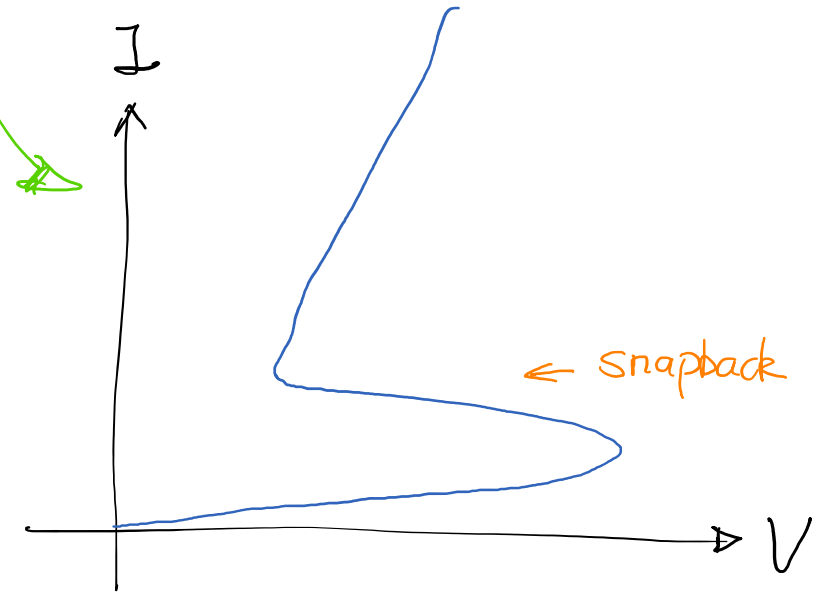
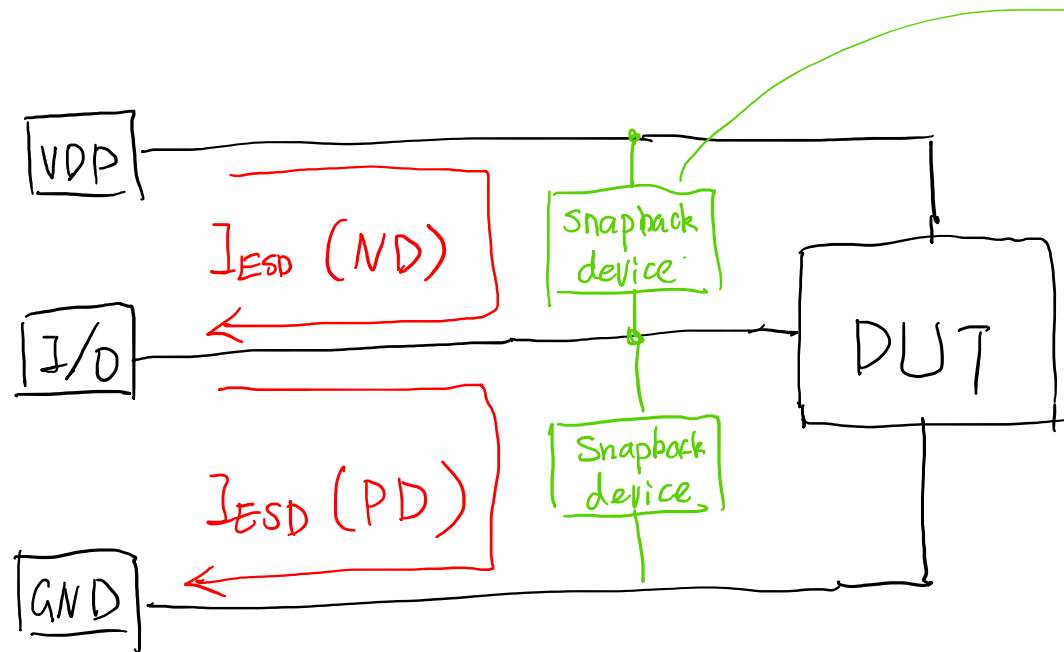
Example



HBM ESD Event



Snapback-Based ESD Protection



Limit the voltage for
ESD protection.



- Next Lecture, we will dive into layout design